Project Report
on

Reflection Groups and Coxeter Groups

Submitted by

Megha Singla

Under the supervision of

Prof. Shripad M. Garge

Department of Mathematics

Indian Institute of Technology Bombay



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INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
DEPARTMENT OF MATHEMATICS

CERTIFICATE

This is to certify that **Megha Singla** has successfully completed the project entitled **"Reflection Groups and Coxeter Groups"** under my supervision as part of the Research Internship in Mathematics at the Department of Mathematics, Indian Institute of Technology Bombay. This report presents the work carried out by her during the internship and is submitted in fulfillment of the requirements of the internship programme.

Prof. Shripad M. Garge
Department of Mathematics
Indian Institute of Technology Bombay

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Introduction

Symmetry is one of the most fundamental ideas in mathematics. Many geometric objects, algebraic structures, and physical systems possess symmetries, and understanding these symmetries often reveals important information about the objects themselves. Reflection groups form a particularly significant class of symmetry groups. They are generated by reflections across hyperplanes in a Euclidean vector space and arise naturally in geometry, algebra, combinatorics, Lie theory, and mathematical physics. Because of their rich structure and wide range of applications, reflection groups have become a central object of study in modern mathematics.

A reflection is a linear transformation that fixes a hyperplane pointwise and reverses the direction orthogonal to that hyperplane. Although individual reflections are simple geometric transformations, groups generated by them exhibit remarkably intricate algebraic and geometric properties. Finite reflection groups provide a natural setting in which geometry and algebra interact closely. The geometry of reflecting hyperplanes gives rise to chambers, root systems, and fundamental domains, while the algebraic structure of the group is encoded through Coxeter systems and Coxeter graphs.

One of the first major achievements in the theory of reflection groups is their classification. Coxeter showed that finite reflection groups can be described using Coxeter diagrams and that irreducible finite reflection groups belong to a small collection of families together with a few exceptional cases. This classification provides a complete understanding of the possible structures that finite reflection groups can possess. Associated with these groups are root systems, which offer a geometric realization of the underlying symmetries and play a fundamental role in the study of Lie algebras and Weyl groups.

Among finite reflection groups, crystallographic reflection groups occupy a special position. These are the reflection groups that preserve a lattice and arise naturally in the theory of semisimple Lie algebras. The corresponding root systems satisfy an additional integrality condition and are known as crystallographic root systems. The groups generated by these root systems

are called Weyl groups. Weyl groups not only describe symmetries of root systems but also govern many structural properties of Lie algebras, algebraic groups, and related mathematical objects.

Beyond their geometric and combinatorial aspects, reflection groups possess a rich invariant theory. Given a reflection group acting on a vector space, one may study the polynomial functions that remain unchanged under the action of the group. The collection of such invariant polynomials forms a graded algebra whose structure reflects important properties of the group. A remarkable result of Chevalley, Shephard, and Todd states that for finite reflection groups, the invariant algebra is itself a polynomial algebra generated by algebraically independent homogeneous invariants. The degrees of these basic invariants provide significant numerical information about the reflection group.

Another important collection of numerical invariants associated with a reflection group is given by its exponents. These numbers arise naturally from the action of Coxeter elements and are closely connected to the geometry of root systems and the combinatorics of Coxeter groups. One of the striking features of the theory is the relationship between the degrees of the basic invariant polynomials and the exponents of the group. This connection reveals a deep interaction between invariant theory, geometry, and group structure.

The purpose of this report is to develop the theory of finite reflection groups and to explore the interplay between their geometric, algebraic, and invariant-theoretic aspects. We begin with the basic definitions of reflections, reflection groups, Coxeter systems, and root systems, followed by the classification of finite reflection groups through Coxeter diagrams. We then study crystallographic root systems, Weyl groups, fundamental domains, and the construction of classical root systems. Finally, we investigate the invariant theory of reflection groups, the role of Coxeter elements and exponents, and the relationship between exponents and the degrees of basic invariants. Through these topics, the report aims to provide a coherent introduction to the rich structure and significance of finite reflection groups.

Chapter 1

Introduction to Reflection Groups

Let W be a finite reflection group acting on a Euclidean space V. Since W is generated by reflections, understanding the structure of W requires a detailed study of these reflections and the geometric objects associated with them. Each reflection is determined by a root, and the collection of all roots forms a root system Φ. The theory of root systems provides a convenient framework for studying reflection groups by translating questions about group elements into questions about vectors in V.

In Section 1.1, we introduce root systems and show that every root system contains a distinguished subset called a *simple system* Δ. This simple system serves as a basis for the span of Φ and allows every root to be expressed as an integral linear combination of simple roots with coefficients of the same sign. Moreover, the reflections corresponding to the simple roots generate the entire reflection group W. Thus, the study of W can be reduced to the study of its simple roots and simple reflections.

Section 1.2 focuses on the relations satisfied by these generating reflections. We show that the simple reflections satisfy certain relations known as the *Coxeter relations*. As a consequence, every finite reflection group can be realized as a Coxeter group.

Finally, in Section 1.3, we classify finite reflection groups. The relations among simple reflections are encoded in a *Coxeter graph*, whose vertices correspond to simple reflections and whose edges record the orders of products of generators. By determining all possible connected Coxeter graphs of positive definite type, we obtain a complete classification of irreducible finite reflection groups. This classification forms the foundation for the study of root systems, Weyl groups, and invariant theory in the subsequent chapters.

1.1 Reflections and the associated Root Systems

Definition 1.1.1 (Reflection)**.** *Let V be a finite-dimensional Euclidean Space. A linear operator s on V is said to be a reflection if there exists $0 \neq \alpha \in V$ such that s sends α to $-\alpha$ and fixes pointwise the hyperplane H_α orthogonal to α.*

Mathematically, it is given by the formula:

$$s(\lambda) = s_\alpha(\lambda) = \lambda - 2\frac{\langle \alpha, \lambda \rangle}{\langle \alpha, \alpha \rangle}\alpha$$

Properties of a reflection

1. The map s_α is an orthogonal transformation, i.e., it preserves the inner product on V.

2. The map s_α has order 2 in the group of all orthogonal transformations on V.

Definition 1.1.2 (Finite Reflection Group)**.** *A finite subgroup of $GL(V)$, where V is a Euclidean space, generated by reflections, is called a Finite Reflection Group. We denote it by W.*

Note that, when we say that W is a reflection group, then it is understood that it is acting on some Euclidean space V and W is generated by reflections in $GL(V)$.

Example 1.1.1. *1. $I_2(m)$: The dihedral group of order 2m acting on the Euclidean plane $(\mathbb{R}^2)$.*

2. $A_n, n \geq 1$: The symmetric group S_{n+1} acting on $V = \{\mathbb{R}^{n+1} | a_1 + a_2 + \cdots + a_{n+1} = 0\}$ by permuting the standard basis of $\mathbb{R}^{n+1}$.
Note that the standard basis of $\mathbb{R}^{n+1}$ is not lying inside V and thus it is first defined as S_{n+1} acting on $\mathbb{R}^{n+1}$ and then restricting the action on V to make W essential.

3. B_n, $n \geq 2$: The semidirect product of S_n and the group of sign changes acting on $\mathbb{R}^n$. This group is naturally isomorphic to the wreath product, $\mathbb{Z}/2\mathbb{Z} \wr S_n$.

4. D_n, $\boldsymbol{n \geq 4}$:The semidirect product of S_n and the group of even number of sign changes, which is a subgroup of B_n of index 2, acting on $\mathbb{R}^n$.

Definition 1.1.3 (Essential reflection group)**.** *A reflection group W is said to be an essential reflection group if there is no non-zero element of V which is fixed by the whole of W.*

Now, our main goal is to classify all finite reflection groups, and thus, in doing so, we first explore the way in which it acts on V. Each reflection s_α in W determines a line L_α and the hyperplane H_α orthogonal to α.

We first see that each $w \in W$ permutes the finite collection of all such lines. Also, if we take the collection of all unit vectors lying on these lines, then w permutes such a set.

Proposition 1.1.4. *Let W be a finite reflection group, and let $w, s_\alpha \in W$. Then $s_{w\alpha} = ws_\alpha w^{-1}$ and hence $s_{w\alpha} \in W$.*

Proof. Let x be an arbitrary element of V. Then,

$$ws_\alpha w^{-1}(x) = w(s_\alpha w^{-1}(x)) = w\left(w^{-1}(x) - 2\frac{\langle \alpha, w^{-1}(x)\rangle}{\langle \alpha, \alpha\rangle}\alpha\right) = x - 2\frac{\langle w\alpha, x\rangle}{\langle w\alpha, w\alpha\rangle}w\alpha = s_{w\alpha}(x)$$

Hence, $s_{w\alpha} \in W$, whenever $w, s_\alpha \in W$. □

Definition 1.1.5 (Root System)**.** *A finite set $\Phi \subset V$ is said to be a root system if it satisfies the following 2 conditions:*

1. *$L_\alpha \cap \Phi = \{\alpha, -\alpha\}$ for all $\alpha \in \Phi$.*
2. *$s_\alpha(\Phi) = \Phi$ for all $\alpha \in \Phi$.*

Note that one can prove that the group generated by the reflections corresponding to elements of a root system is always finite.

Theorem 1.1.6. *Let Φ be a finite subset of V satisfying the following conditions:*

1. *$L_\alpha \cap \Phi = \{\alpha, -\alpha\}$ for all $\alpha \in \Phi$.*
2. *$s_\alpha(\Phi) = \Phi$ for all $\alpha \in \Phi$.*

The group W generated by $\{s_\alpha : \alpha \in \Phi\}$ is finite.

Proof. Let $\rho : W \to Sym(\Phi)$ be the natural homomorphism defined by $\rho(w)(\alpha) = w\alpha$ for all $\alpha \in \Phi$ and all $w \in W$. The homomorphism is well defined because of property 2 of Φ. We intend to prove that ρ is injective. Let $w \in \ker(\rho)$, which implies $\rho(w)(\alpha) = \alpha$ for all $\alpha \in \Phi$. It follows that w preserves the spanning set of Φ pointwise. Also, each $w_0 \in W$ preserves the orthogonal complement of span(Φ) pointwise. So, $w \in \ker(\rho)$ implies that $w = I$, and hence ρ is injective. Since Φ is finite, it follows that Sym(Φ) is finite, and hence W is finite. □

Also, from a finite reflection group we can construct Φ. So, we now try to study W with the help of this root system Φ. However, there is one serious drawback of using Φ for our purpose, that is its cardinality. So, we want to study a smaller subset of Φ from which Φ can be reconstituted. This leads us in search of a new system which we call the **Simple System**.

Definition 1.1.7 (Simple System)**.** *Let Δ be a subset of Φ. If Δ is a basis for the linear span of Φ and if every $\alpha \in \Phi$ is a real linear combination of elements of Δ with all coefficients having the same sign, then we say that Δ is a simple system for Φ.*

The condition of like sign is useful because of the geometric condition that if we take any 2 distinct members of Δ, then their inner product is ≤ 0, which will later help us in classifying all the reflection groups.

Also, the existence of a simple system leads to a partition of Φ into 2 disjoint sets Π, known as the positive system, consisting of all those members of Φ which are written in the members of Δ with all coefficients having non negative sign and $-\Pi$, the negative system, consisting of all those members of Φ which are written in the members of Δ with all coefficients having non positive sign.
It is not at all evident that a simple system exists, however, it can be proven with the help of Π, which can be obtained with respect to some ordered basis and lexicographic ordering.

Theorem 1.1.8. *Given a positive system Π, there exists a unique simple system Δ contained in it, i.e., a simple system exists. Also, a given simple system leads to a unique positive system.*

We do not prove the theorem here.

Corollary 1.1.9. *Given a simple system Δ, if we take any 2 distinct members of Δ, then their inner product is ≤ 0.*

Proof. Let α and β be 2 distinct members of Δ such that $\langle \alpha, \beta \rangle > 0$. Then,

$$s_\alpha(\beta) = \beta - 2\frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}\alpha$$

implies that $s_\alpha(\beta)$ is written as a linear combination of the members of Δ, one having a positive coefficient and another having a negative coefficient. **Since Δ is a basis, there is a unique expression for elements in its span.** this is a contradiction to the fact that every member of Φ is written as a linear combination of the members of Δ, all having the same sign. $\square$

Note that a positive system Π is with respect to some basis, and thus the positive system (respectively, a simple system) is not unique.

Proposition 1.1.10. *Let Δ be the simple system contained in Π, and $\alpha \in \Delta$. Then $s_\alpha(\Pi \setminus \{\alpha\}) = \Pi \setminus \{\alpha\}$.*

Proof. Let $\beta \in \Pi \setminus \{\alpha\}$, then

$$s_\alpha(\beta) = \beta - 2\frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}\alpha$$

The positive coefficient of β implies that $s_\alpha(\beta) \in \Pi$. Also, $s_\alpha(\beta) \neq \alpha$ because if it were so, then $\beta = s_\alpha(\alpha) = -\alpha$, contradicting the fact that $\beta \in \Pi$. □

Theorem 1.1.11. *Let Π and Π' be 2 positive systems. Then there exists $w \in W$ such that $w\Pi = \Pi'$.*

Proof. If $\Pi = \Pi'$, then $w = I$ will work. Let $\Pi \neq \Pi'$, then $|\Pi \cap -\Pi'| \neq 0$, which implies that $|\Delta \cap -\Pi'| \neq 0$ (if $|\Delta \cap -\Pi'| = 0$, then $\Delta \subset \Pi'$. Since Δ determines the positive system uniquely, this implies that $\Pi = \Pi'$.) Let $\alpha \in \Delta \cap -\Pi'$ and $|\Pi \cap -\Pi'| = r$. Then $|s_\alpha \Pi \cap -\Pi'| = r - 1$, because now α will not contribute to the intersection, whereas it was previously contributing. By induction, we can get w such that $|w\Pi \cap -\Pi'| = 0$, which implies that $w(\Pi) = \Pi'$. □

Definition 1.1.12 (Height). *If $\beta \in \Phi$, then*

$$\beta = \sum_{\alpha \in \Delta} c_\alpha \alpha.$$

We define $\sum c_\alpha$ to be the height of β.

Theorem 1.1.13. *For a fixed simple system Δ, W is generated by simple reflections.*

Let $W' = \langle s_\alpha \mid \alpha \in \Delta \rangle$. We intend to prove that $W = W'$. We will prove this in 3 steps. First, we will prove that the orbit of every $\beta \in \Pi$ (w.r.t. W') contains a member of Δ. Choose $\alpha \in W'\beta \cap \Pi$ of smallest height. We claim that $\alpha \in \Delta$. Let $\alpha \notin \Delta$, then $\alpha = \sum_{\gamma \in \Delta} c_\gamma \gamma$. Since $0 < \langle \alpha, \alpha \rangle = \langle \alpha, \sum c_\gamma \gamma \rangle = \sum c_\gamma \langle \alpha, \gamma \rangle$, there exists some γ_0 such that $\langle \alpha, \gamma \rangle > 0$. If $\gamma_0 = \alpha$, we are done. Otherwise, consider

$$s_{\gamma_0}(\alpha) = \alpha - 2\frac{\langle \gamma_0, \alpha \rangle}{\langle \gamma_0, \gamma_0 \rangle}\gamma_0.$$

Clearly, $s_{\gamma_0}(\alpha) \in \Pi$, since s_{γ_0} will preserve $\Pi \setminus \{\gamma_0\}$. Also, $s_{\gamma_0}(\alpha) \in W'\beta$, as $\alpha \in W'\beta$ and $s_{\gamma_0} \in W'$. But the height of $s_{\gamma_0}(\alpha)$ is less than the height of α, because $\langle \alpha, \gamma \rangle > 0$, contradicting the choice of α.
Next, we intend to prove that $W'\Delta = \Phi$. Since W preserves Φ, $W'\Delta \subset \Phi$ is trivial. So, we only need to prove that $\Phi \subset W'\Delta$. Let $\beta \in \Pi$. From step 1, there exist $w' \in W'$ and $\alpha \in \Delta$ such that $w'\beta = \alpha$, which implies that $\beta = (w')^{-1}\alpha$, and hence $\beta \in W'\Delta$. If $\beta \in -\Pi$, then $-\beta \in \Pi$, so there exist $w' \in W'$ and $\alpha \in \Delta$ such that $w'(-\beta) = \alpha$, which implies that $\beta = w'(s_\alpha(\alpha)) = (w's_\alpha)(\alpha)$, and hence $\beta \in W'\Delta$. It follows that $\Phi \subset W'\Delta$, and hence $W'\Delta = \Phi$.
Now, in order to prove that $W' = W$, we take any arbitrary reflection s_α for $\alpha \in \Phi$ from W and prove that it is in W'. Let $s_\alpha \in W$ for $\alpha \in \Phi$. From step 2, there exist $w \in W'$ and $\beta \in \Delta$ such that $w\beta = \alpha$. Consider $s_\alpha = s_{w\beta} = ws_\beta w^{-1}$. Since $w \in W'$ and $s_\beta \in W'$, it follows that $s_\alpha \in W'$. Hence, each reflection s_α for $\alpha \in \Phi$ belongs to W', which implies that $W = W'$.

1.2 Relation satisfying by simple reflections

Definition 1.2.1 (Length)**.** *Let W be a finite reflection group. Then the length of $w \in W$ is the smallest r for which the expression of w in terms of simple reflections exists, denoted by $l(w)$, and the corresponding expression is called* ***reduced****. Note that, by hypothesis, $l(1) = 0$.*

Properties of length function

1. $l(w) = 1$ if and only if $w = s_\alpha$, for some $\alpha \in \Delta$.
2. $l(w) = l(w^{-1})$.
3. $\det(w) = (-1)^{l(w)}$.
4. $l(ww')$ has the same parity as that of $l(w) + l(w')$.

Definition 1.2.2. *$n(w)$ is the number of positive roots sent to negative roots by w, i.e., $n(w) = card(\Pi \cap w^{-1}(-\Pi))$.*

We will next characterise the length of w by $n(w)$, i.e., the number of positive roots sent to negative roots by w, after establishing some lemmas and corollaries.

Lemma 1.2.3. *Let $\alpha \in \Delta$. Then,*

1. *If $w\alpha > 0$, then $n(ws_\alpha) = n(w) + 1$. This follows from the fact that s_α preserves $\Pi \setminus \{\alpha\}$.*

2. *If* $w\alpha < 0$, *then* $n(ws_\alpha) = n(w) - 1$.

Replacing w *by* w^{-1} *in the above, and using the fact that* $n(w) = n(w^{-1})$, *we get,*

1. *If* $w^{-1}\alpha > 0$, *then* $n(s_\alpha w) = n(w) + 1$.
2. *If* $w^{-1}\alpha < 0$, *then* $n(s_\alpha w) = n(w) - 1$.

Corollary 1.2.4. *If* $w \in W$ *is written in terms of simple reflections as* $w = s_1 s_2 \cdots s_r$, *then* $n(w) \leq r$.

Theorem 1.2.5 (**Deletion Condition**). *Given a simple system* Δ *and the corresponding positive system* Π, *if* $w = s_1 s_2 \cdots s_r$ *is the expression of* w *in terms of simple reflections and* $n(w) < r$, *then there exist indices* i *and* j *such that* $w = s_1 s_2 \cdots \hat{s_i} s_{i+1} \cdots \hat{s_j} \cdots s_r$, *where the hat denotes omission.*

Corollary 1.2.6. *From the deletion condition, we can conclude that* $n(w) = l(w)$.

Theorem 1.2.7 (**Exchange Condition**). *If* $w = s_1 s_2 \cdots s_r$, *then* w *has a reduced expression in terms of simple reflections ending with* s *if and only if* $l(ws) < l(w)$.
In other words, if $w = s_1 s_2 \cdots s_r$, *then there exists an index* i *such that* $w = s_1 s_2 \cdots \hat{s_i} \cdots s_r s$ *if and only if* $l(ws) < l(w)$.
Note that if $w = s_1 s_2 \cdots s_r$ *is the reduced expression of* w, *then* s_i *is uniquely determined.*

From Corollary 1.2.6, we can conclude the following theorem, which states that if $w \in W$ sends no positive root to a negative root, then it is the identity.

Theorem 1.2.8. *Let* Δ *be the simple system and* Π *be the corresponding positive system. Then the following are equivalent:*

1. $w(\Delta) = \Delta$
2. $w(\Pi) = \Pi$
3. $n(w) = 0$
4. $l(w) = 0$
5. $w = I$

Also, there exists a unique $w_0 \in W$ such that $w_0(\Pi) = -\Pi$.

Now, if $w_0(\Pi) = -\Pi$, then $w_0^{-1}(\Pi) = -\Pi$ (since $n(w) = n(w_0)$), which implies $w_0 = w_0^{-1}$.

Characterisation of w_0: If $w \in W$ is such that $l(ws_\alpha) < l(w)$ for all $\alpha \in \Delta$, then $w = w_0$.

Also note that if $w_1 = s_1 s_2 \cdots s_l$ is the reduced expression of w_1, then there exists $w_2 \in W$ such that $w_0 = w_1 w_2$ and $l(w_0) = l(w_1) + l(w_2)$.

$$l(w_0) - l(w_1) = l(w_2) = l(w_1^{-1} w_0) = l(w_0^{-1} w_1) = l(w_0 w_1) \quad \forall\, w_1 \in W$$

which can be restated as follows,

$$l(w_0 w) = l(w_0) - l(w) \quad \forall\, w \in W$$

Theorem 1.2.9. *Let Δ be the simple system. Then W is generated by simple reflections, subject only to the relations*

$$(s_\alpha s_\beta)^{m(\alpha,\beta)} = 1$$

where $m(\alpha, \beta)$ denotes the order of the element $(s_\alpha s_\beta)$.

Definition 1.2.10 (**Coxeter Group**)**.** *W is said to be a Coxeter group if there exists $S \subset W$ such that W can be generated by the members of S, subject only to the relations $(s_\alpha s_\beta)^{m(\alpha,\beta)} = 1$ for all $s_\alpha, s_\beta \in S$, and the pair (W, S) is called a* ***Coxeter System****.*

It directly follows that every finite reflection group is a Coxeter group.

1.3 Coxeter Graphs and the Classification of Finite Reflection Groups

Definition 1.3.1 (Coxeter Graph)**.** *Let Δ be the simple system. Then the graph in which $\alpha, \beta \in \Delta$ are joined by an edge if and only if $m(\alpha, \beta) \geq 3$, where $m(\alpha, \beta)$ is the order of $s_\alpha s_\beta$, with the edge labelled by $m(\alpha, \beta)$ if $m(\alpha, \beta) \geq 4$, left unlabelled when $m(\alpha, \beta) = 3$, and not joined when $m(\alpha, \beta) = 2$, is known as a Coxeter graph, denoted by Γ.*

Since every finite reflection group is a Coxeter group, this Coxeter graph encodes all essential relations of simple reflections.

Proposition 1.3.2. *Let W_1 and W_2 be 2 finite essential reflection groups acting on V_1 and V_2 respectively, such that both have the same Coxeter graph. Then they are isomorphic. In particular, if $V_1 = V_2$, then W_1 and W_2 are conjugate in $O(V)$.*

Proof. We have W_1 and W_2 acting on V_1 and V_2 essentially, respectively. Hence $\dim(V_1) = \dim(V_2)$ and the cardinalities of Δ_1 and Δ_2 are the same. Since both groups have the same Coxeter graph, there exists a map $\phi : V_1 \hookrightarrow V_2$ such that $\phi(\Delta_1) = \Delta_2$ and $m(\alpha, \beta) = m(\phi(\alpha), \phi(\beta))$ for all $\alpha, \beta \in \Delta_1$. Also, we can assume that every member of Δ_1 and Δ_2 has unit length. Since the inner product of α, β is completely dependent on $m(\alpha, \beta)$, this ϕ is inner product preserving and norm preserving, and thus an isometry. Now note that $\phi s_\alpha \phi^{-1}(x) = \phi(\phi^{-1}(x) - \langle \phi^{-1}(x), \alpha \rangle \alpha) = x - \langle x, \phi(\alpha) \rangle \phi(\alpha) = s_{\phi(\alpha)}(x)$. Define a map $\chi : W_1 \to W_2$ such that $\chi(s_\alpha) = \phi s_\alpha \phi^{-1} = s_{\phi(\alpha)}$. Clearly, χ is bijective as well as a homomorphism. Thus W_1 and W_2 are isomorphic. In particular, if $V_1 = V_2$, then ϕ is from V_1 to V_2, and hence W_1 and W_2 are conjugate. □

Definition 1.3.3 (Connected Coxeter Graph)**.** *A Coxeter graph is said to be connected if for any two elements of Δ, there exists a path joining them, and the corresponding Coxeter system (W, S) is said to be* ***irreducible****.*

Let Γ_1, Γ_2, $\cdots$, Γ_r be the connected components of Γ. Then 2 distinct members α, β of Δ coming from 2 different connected components will commute, since $m(\alpha, \beta) = 2$ implies $s_\alpha s_\beta s_\alpha s_\beta = 1$, which implies $s_\alpha s_\beta = s_\beta s_\alpha$.

Proposition 1.3.4. *Let (W, S) be a Coxeter system and let Γ be its Coxeter graph. Suppose that $\Gamma_1, \Gamma_2, \ldots, \Gamma_r$ are the connected components of Γ. Let $S = S_1 \sqcup S_2 \sqcup \cdots \sqcup S_r$ be the corresponding partition of S. Let $W_i = \langle S_i \rangle$ be the parabolic subgroup of W generated by S_i. Then $W = W_1 \times W_2 \times \cdots \times W_r$.*

Proof. In order to prove that $W = W_1 \times W_2 \times \cdots \times W_r$, we need to prove 3 things: that each indicated parabolic subgroup is normal in W, that their pairwise intersection is trivial, and that together they form the whole group. As shown above, s_α and s_β commute for all $s_\alpha \in W_i$ and $s_\beta \in W_j$ for $i \neq j$, and thus the centralizer of each parabolic subgroup is W itself, indicating that each one is normal in W. Also, we know that $W_{I \cap J} = W_I \cap W_J$. Since $\Delta_i \cap \Delta_j = \emptyset$, it follows that $W_I \cap W_J$ is trivial. Hence $W = W_1 \times W_2 \times \cdots \times W_r$. □

Next, we see that to each Coxeter graph Γ, there is an associated symmetric $n \times n$ matrix A. The entries of A are given by

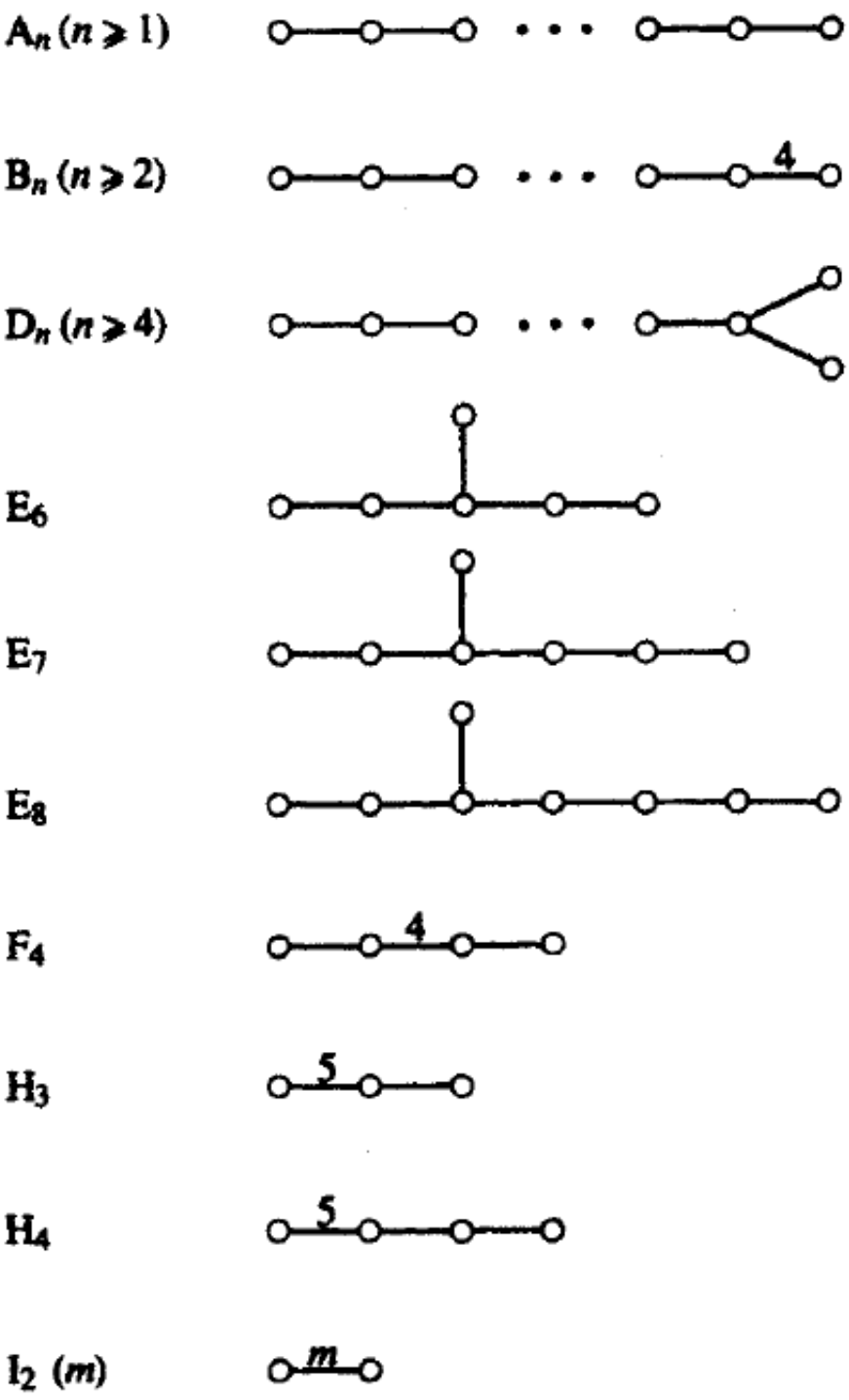
$$a_{ss'} = -\cos\left(\frac{\pi}{m(s, s')}\right),$$

where $m(s, s')$ is the order of the element ss'. Moreover, corresponding to each symmetric matrix, there exists an associated bilinear form ($\langle x, y \rangle = x^T A y$) and a quadratic form.

Definition 1.3.5 (Positive Definite Matrix)**.** *A matrix A is said to be positive definite if $\langle x, x\rangle = x^T Ax > 0$ for all $x \neq 0$.*

Next, we conclude that every matrix A arising from a finite reflection group W is positive definite. This is because the entries of the matrix are the inner products of the corresponding linearly independent vectors.
Thus, our strategy for classifying finite reflection groups is as follows: first, we list all possible connected positive definite Coxeter graphs, and then show that every connected positive definite Coxeter graph corresponds to a finite reflection group.

Example 1.3.1 (Examples of Connected Positive Definite Graphs)**.** A_n $(n \geq 1)$, B_n $(n \geq 2)$, D_n $(n \geq 4)$, E_6, E_7, E_8, F_4, H_3, H_4, $I_2(m)$.



To see that these graphs are positive definite, it suffices to compute their

determinants. This follows from an **inductive computation of determinants**.

More precisely, to prove that a Coxeter graph with n vertices is positive definite, it is enough to show that its determinant is positive, since all its principal minors are positive by the inductive assumption that every Coxeter graph with fewer than n vertices is positive definite.
Since the label 3 appears repeatedly on the edges of Coxeter graphs, the associated matrix contains the entry $-\frac{1}{2}$ many times. Therefore, it is convenient to compute the determinant of the matrix $2A$ instead of A.

Next, we have a formula to compute the determinant of the matrix $2A$,

where A is the matrix associated with a Coxeter graph. This formula applies in the situation observed in our examples. In these examples, the Coxeter graphs satisfy the following observation:

- The last vertex, numbered n, is joined by an edge to exactly one other vertex, namely the vertex numbered $n-1$. Moreover, the edge joining vertices $n-1$ and n is labelled $m=3$ or $m=4$.

Under this observation, the determinant of $2A$ satisfies the formula

$$\det(2A) = 2d_{n-1} - c\, d_{n-2},$$

where d_i denotes the determinant of the upper-left $i \times i$ submatrix of $2A$.
Using the above formula, we can compute the determinant of each matrix, and see that each one is positive, and hence each graph is positive definite.
The determinant of $2A$ corresponding to $\mathbf{A_n}$, denoted by d_n, can be computed inductively.
The determinant of $2A$ corresponding to $\mathbf{A_1}$, denoted by d_1, is 2.
The determinant of $2A$ corresponding to $\mathbf{A_2}$, denoted by d_2, is 3.
$d_3 = 2(3) - 2 = 4$
...
$d_{n-2} = 2d_{n-3} - d_{n-4}$
$d_{n-1} = 2d_{n-2} - d_{n-3}$
$d_n = 2d_{n-1} - d_{n-2}$
Adding all the equations, we get: $d_n = d_{n-1} + 1$
and thus $d_n = n + 1$.
The determinant of $2A$ corresponding to $\mathbf{B_n}$ is $2(d_{n-1}) - 2(d_{n-2}) = 2(n) - 2(n-1) = 2$.
The determinant of $2A$ corresponding to $\mathbf{D_n}$ is $2(d_{n-1}) - 2d_{n-3} = 2(n) - 2(n-2) = 4$.
The determinant of $2A$ corresponding to $\mathbf{E_6}$ is $2(|D_5|) - |A_4| = 2(4) - 5 = 3$.

The determinant of $2A$ corresponding to $\mathbf{E_7}$ is $2|E_6| - |D_5| = 2(3) - 4 = 2$.
The determinant of $2A$ corresponding to $\mathbf{E_8}$ is $2|E_7| - |E_6| = 2(2) - 3 = 1$.
The determinant of $2A$ corresponding to $\mathbf{F_4}$ is $3|B_3| - |A_2| = 2(2) - 3 = 1$.
The determinant of $2A$ corresponding to $\mathbf{H_3}$:

$$2H_3 = \begin{pmatrix} 2 & -\frac{1+\sqrt{5}}{2} & 0 \\ -\frac{1+\sqrt{5}}{2} & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

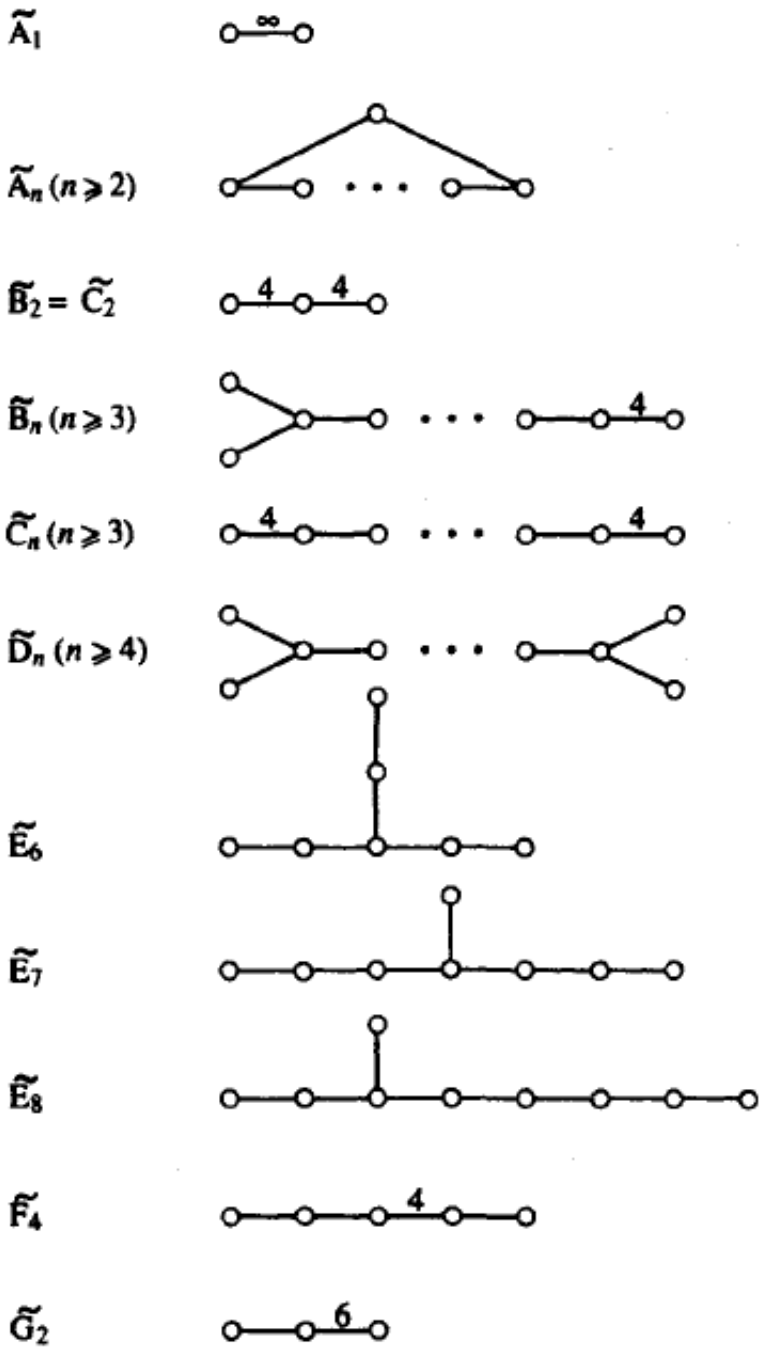
$$\det(2H_3) = 3 - \sqrt{5}.$$

The determinant of $2A$ corresponding to $\mathbf{I_2(m)}$:

$$2I_2(m) = \begin{pmatrix} 2 & -2\cos\left(\frac{\pi}{m}\right) \\ -2\cos\left(\frac{\pi}{m}\right) & 2 \end{pmatrix}.$$

$$\det\big(2I_2(m)\big) = 4 - 4\cos^2\left(\frac{\pi}{m}\right) = 4\sin^2\left(\frac{\pi}{m}\right).$$

Next, our aim is to prove that the examples discussed above are the only connected positive definite graphs.
Now, we list some positive semidefinite graphs which are not positive definite: $\tilde{A}_1, \tilde{A}_n, \tilde{B}_n, \tilde{C}_n, \tilde{D}_n, \tilde{E}_6, \tilde{E}_7, \tilde{E}_8, \tilde{F}_4, \tilde{G}_2$.



Listing these graphs will help us in classifying all the finite reflection groups.

Definition 1.3.6 (Subgraph)**.** *A graph obtained by removing some vertices, decreasing the label on the edges, or both, is known as a subgraph.*

We claim that every connected Coxeter graph of positive definite type can have only positive definite subgraphs. Thus, the graphs of positive semidefinite type listed above cannot be subgraphs of any connected positive definite Coxeter graph. This will be a corollary of the proposition we now discuss.

Proposition 1.3.7. *Let A be a real, symmetric $n \times n$ matrix of positive type which is indecomposable such that $a_{ij} \leq 0$ for all $i \neq j$. Then,*

1. *The null space $N = \{x \in \mathbb{R}^n \mid x^T A x = 0\}$ is the null space of A and has dimension less than or equal to 1.*

2. *The smallest eigenvalue of A has multiplicity 1 and has an eigenvector whose all coordinates are strictly positive.*

Proof. 1. It is clear that the null space of A is contained in N. Conversely, let $x \in N$; we intend to prove that $Ax = 0$.
Since A is a real and symmetric matrix, it is orthogonally diagonalisable. So, there exists P, an orthogonal matrix, such that $P^T AP = D$. Since P is invertible, there exists y such that $Py = x$.
We have $x^T Ax = 0 \implies (Py)^T A(Py) = 0 \implies (y^T P^T)A(Py) = y^T Dy = 0 \implies \sum d_i y_i^2 = 0$. Since A is a positive type matrix, $d_i \geq 0$ for all i. It follows that $\sum d_i y_i = 0 \implies Dy = 0 \implies y \in \text{Null}(D) \implies Dy = 0 \implies (P^T AP)(y) = 0 \implies P^{-1}(Ax) = 0 \implies Ax = 0 \implies x \in \text{Null}(A)$.
Now, we intend to prove that the dimension of the null space is less than or equal to 1. Suppose the dimension of the null space is positive. It implies that there exists $0 \neq x \in \mathbb{R}^n$ such that $x^T Ax = 0$. Consider $z = (|x_1|, |x_2|, |x_3|, \cdots, |x_n|)$. We claim that z is also in the null space of A. This is because $z^T Az \leq x^T Ax$, and since A is positive semidefinite, it follows that $0 \leq z^T Az \leq x^T Ax = 0$. We thus have a vector in the null space whose all coordinates are greater than or equal to 0. We claim that all coordinates of z are greater than 0. Since z is in the null space of A, we have $Az = 0 \implies \sum_j a_{ij}|x_j| = 0$ for all i. Let I be the set of indices such that $z_i = 0$ and J be the set of indices such that $z_i \neq 0$. In particular, $\sum_j a_{ij}|x_j| = 0$ for all $j \in J$ and $i \in I$, which implies that $a_{ij} = 0$ for all $i \in I$ and $j \in J$. But then A is decomposable if I is non-empty. Since A is indecomposable, it follows that I is empty, and hence all coordinates of x are non-zero. It now follows from the discussion that any vector in the null space of A must have all coordinates non-zero. If we have 2 linearly independent vectors in the null space, then we can find a linear combination of those 2 vectors such that the resulting vector is non-zero but some coordinate is zero.

But then it cannot be in the null space of A. Hence, the null space of A has dimension less than or equal to 1.

2. Now, we intend to prove that the smallest eigenvalue of A has multiplicity 1 and has an eigenvector whose all coordinates are strictly positive. Since A is real and symmetric, there exists P such that $P^TAP = D$. Let d be the smallest eigenvalue of A. Consider $D - dI$, and note that it satisfies all the assumptions of the proposition. Hence, the zero eigenvalue has exactly multiplicity 1, and has an eigenvector whose all coordinates are strictly positive. It follows that the smallest eigenvalue of A has multiplicity 1 and has an eigenvector whose all coordinates are strictly positive.

□

Corollary 1.3.8. *Let Γ be a connected Coxeter graph of positive type. Then every proper subgraph of Γ is positive definite.*

Proof. Let Γ be a connected Coxeter graph of positive type, let Γ' be its proper subgraph, and let A and A' be the corresponding Coxeter matrices of size $n \times n$ and $k \times k$ respectively. We intend to prove that A' is positive definite. Suppose, if possible, that A' is not positive definite. Then there exists $x \in \mathbb{R}^k$ such that $x^TA'x \leq 0$, where $x = (x_1, x_2, \cdots, x_k)$. Consider $y = (|x_1|, |x_2|, \cdots, |x_k|, 0, 0, \cdots, 0)$. Note that

$$0 \leq y^TAy = \sum a_{ij}|x_i||x_j| \leq \sum a'_{ij}|x_i||x_j| \leq \sum a_{ij}x_ix_j \leq 0.$$

The first inequality holds because A is positive semidefinite, and the second inequality holds because $a_{ij} \leq a'_{ij}$. This is because

$$m_{ij} \geq m'_{ij} \implies \frac{\pi}{m_{ij}} \leq \frac{\pi}{m'_{ij}} \implies \cos\left(\frac{\pi}{m_{ij}}\right) \geq \cos\left(\frac{\pi}{m'_{ij}}\right)$$
$$\implies -\cos\left(\frac{\pi}{m_{ij}}\right) \leq -\cos\left(\frac{\pi}{m'_{ij}}\right) \implies a_{ij} \leq a'_{ij}.$$

The next inequality holds because $a'_{ij} \leq 0$, and the last inequality holds because of our assumption. Thus equality holds throughout. It follows that $y \in N$, which implies all coordinates of y are non-zero, and hence $k = n$. Also, from the second equality, that is $\sum a_{ij}|x_i||x_j| = \sum a'_{ij}|x_i||x_j|$, we see that $a_{ij} = a'_{ij}$, and hence $\Gamma = \Gamma'$, contradicting the assumption that Γ' is a proper subgraph of Γ. □

We rely heavily on the above corollary in order to prove that all the connected positive definite Coxeter graphs listed above are the only connected Coxeter graphs of positive definite type.

Chapter 2

Structure of Crystallographic Reflection Groups

In the previous chapter, finite reflection groups were classified using Coxeter systems and Coxeter graphs. Having obtained a complete list of finite reflection groups, the next natural question is to understand the geometric structures associated with them. The purpose of this chapter is to study these geometric objects and to see how they encode the structure of the corresponding reflection groups.

We begin by introducing fundamental chambers and fundamental domains. These provide a geometric description of the action of a reflection group on the ambient Euclidean space and allow us to choose a unique representative from each orbit. We then study isotropy subgroups and show that they themselves possess the structure of reflection groups.

Next, we focus on crystallographic reflection groups and the associated root systems. These root systems satisfy an additional integrality condition and give rise to Weyl groups, which play a central role in Lie theory and many other areas of mathematics. We investigate the restrictions imposed by the crystallographic condition and derive several important properties of roots, including the possible root lengths and the angles between them.

Finally, using the classification obtained in the previous chapter, we construct the irreducible root systems corresponding to the Weyl groups of types A_n, B_n, C_n, D_n, G_2, F_4, and E_8. We also determine their simple systems, positive systems, and highest roots. These constructions provide explicit geometric realizations of the finite reflection groups and prepare the foundation for the study of invariant theory in the next chapter.

2.1 Fundamental Domains

Let W be the finite reflection group with Π as the positive system and Δ be the corresponding simple system. Define C, known as the **Fundamental Chamber**, as

$$C = \{\lambda \in V \mid \langle \lambda, \alpha_i \rangle > 0 \quad \forall\, \alpha_i \in \Delta\}$$
$$D = \bar{C} = \{\lambda \in V \mid \langle \lambda, \alpha_i \rangle \geq 0 \quad \forall\, \alpha_i \in \Delta\}$$

Definition 2.1.1 (Fundamental Domain:)**.** *A subset D' of V satisfying the following two conditions is called a Fundamental Domain:*

1. *For each $\lambda \in V$, there exists $w \in W$ such that $w\lambda \in D'$.*
2. *No two distinct elements of D' are W-conjugate.*

Next, we will characterise the Fundamental Domain by D.

Lemma 2.1.2. *For each $\lambda \in V$, there exist $\mu \in D$ and $w \in W$ such that $w\lambda = \mu$. Moreover, $\mu - \lambda$ is a non-negative $\mathbb{R}$-linear combination of the members of Δ.*

Proof. Take $\lambda \in V$ and define $A = \{\mu = w\lambda \mid w \in W, \mu \geq \lambda\}$, where $\mu \geq \lambda$ means that $\mu - \lambda$ is written as a linear combination of the members of Δ with all coefficients having non-negative signs. Clearly, A is a poset and also finite. It follows that a maximal element of A exists, say μ_0. We claim that $\mu_0 \in D$. We intend to prove that $\langle \mu_0, \alpha \rangle \geq 0$ for all $\alpha \in \Delta$. Suppose there exists some $\alpha \in \Delta$ such that $\langle \mu_0, \alpha \rangle < 0$. Note that

$$s_\alpha(\mu_0) = \mu_0 - 2\frac{\langle \alpha, \mu_0 \rangle}{\langle \alpha, \alpha \rangle}\alpha$$

is clearly greater than μ_0, since $\langle \mu_0, \alpha \rangle < 0$. We will prove that $s_\alpha(\mu_0) \in A$. This is because $\mu_0 = w\lambda$ for some $w \in W$, which implies $s_\alpha(\mu_0) = (s_\alpha w)\lambda$, thus $s_\alpha(\mu_0)$ is also W-conjugate to λ. Also, $s_\alpha(\mu_0) - \lambda = \mu_0 + c\alpha - \lambda \geq 0$, contradicting the maximality of μ_0. Hence, $\langle \mu_0, \alpha \rangle \geq 0$ for all $\alpha \in \Delta$, which implies $\mu_0 \in D$. □

From the last lemma, it is clear that one part of the existence condition in the definition of a Fundamental Domain is satisfied by D. Now, to see uniqueness, we need to show that no two elements of D are W-conjugate. However, while proving this, we will obtain some sharper information.

Definition 2.1.3 (Isotropy Group:)**.** *Let μ be an arbitrary element of V. The isotropy group of μ is defined as*

$$\{w \in W \mid w\mu = \mu\}.$$

Theorem 2.1.4. *Let Π be the positive system and Δ be the corresponding simple system, and define D as above. Then*

1. *No two elements in D are W-conjugate, that is, if $w\lambda = \mu$ for $\lambda, \mu \in D$, then $\lambda = \mu$. Also, if $\lambda \in C$, then the isotropy group of λ is trivial. Hence, D is the fundamental domain of V.*

2. *Any isotropy group of W is itself a reflection group.*

3. *If U is any subset of V, then the subgroup of W fixing U pointwise is also a reflection group.*

Proof. 1. Suppose $w\lambda = \mu$ where $\lambda, \mu \in D$ and $w \in W$. If $w = e$, the identity element, then clearly $\lambda = \mu$, and we are done. Now assume $w \neq e$. Then $n(w) > 0$. Hence there exists $\alpha \in \Delta$ such that $w\alpha < 0$. Thus, $n(ws_\alpha) = n(w) - 1$. We claim that ws_α also sends λ to μ. For this, it is enough to show that s_α fixes λ, i.e., $s_\alpha(\lambda) = \lambda$, which holds if and only if $\langle \alpha, \lambda \rangle = 0$. Since $\lambda \in D$ and $\alpha \in \Delta$, we have $\langle \alpha, \lambda \rangle \geq 0$. We now prove the reverse inequality. Using $w\lambda = \mu$ and W-invariance of the inner product, $\langle \alpha, \lambda \rangle = \langle w\alpha, w\lambda \rangle = \langle w\alpha, \mu \rangle$. Since $w\alpha < 0$, we can write

$$w\alpha = \sum_{\gamma \in \Delta} c_\gamma \gamma \quad \text{with } c_\gamma \leq 0.$$

Therefore,

$$\langle w\alpha, \mu \rangle = \sum_{\gamma \in \Delta} c_\gamma \langle \gamma, \mu \rangle \leq 0,$$

because $\mu \in D$ implies $\langle \gamma, \mu \rangle \geq 0$ for all $\gamma \in \Delta$.

Hence, $\langle \alpha, \lambda \rangle \leq 0$. Combining both inequalities gives $\langle \alpha, \lambda \rangle = 0$, so $s_\alpha(\lambda) = \lambda$. Thus, $ws_\alpha(\lambda) = w(\lambda) = \mu$, and $n(ws_\alpha) = n(w) - 1$. By induction on $n(w)$, we conclude that $\lambda = \mu$. Also, it follows that w is the product of simple reflections fixing λ.

2. Let $\lambda \in V$ and consider $\{w \in W \mid w\lambda = \lambda\}$. From part (1), it follows that there exist $w \in W$ and $\mu \in D$ such that $w\lambda = \mu$. We know the isotropy group of μ, say W', is generated by those reflections which it contains. Now, $W'(\mu) = \mu$ if and only if $W'(w\lambda) = w\lambda$, if and only if $(w^{-1}W'w)(\lambda) = \lambda$. Also note that if s_α is a simple reflection, then $w^{-1}s_\alpha w$ is also a reflection. Thus the isotropy group of λ is generated by those reflections which it contains.

3. Let $\{\alpha_1, \alpha_2, \cdots, \alpha_r\}$ be a basis of span(U). In order to find the subgroup of W which fixes U pointwise, we need to find the subgroup

which fixes this basis pointwise, that is, $W' = \{w \in W \mid w(\alpha_i) = \alpha_i \ \forall \ i \in \{1, \cdots, r\}\}$. If $r = 1$, then we are done, but if $r > 1$, then we need to find $W' = \{w \in W_1 \mid w(\alpha_i) = \alpha_i \ \forall \ i \in \{2, \cdots, r\}\}$. By the induction hypothesis, W' is the subgroup of W generated by those reflections which it contains. Hence, the subgroup of W which fixes U pointwise is generated by those reflections which it contains.

□

We know that $w \in W$ sends Δ to $w\Delta$, which is again a simple system corresponding to the positive system $w\Pi$. Also, $w \in W$ sends C to wC.

2.2 Crystallographic Groups and Weyl groups

Definition 2.2.1 (Crystallographic Group)**.** *A subgroup G of $GL(V)$ is said to be crystallographic if it preserves a lattice, i.e., a $\mathbb{Z}$-linear combination of some basis.*

It is clear from the definition that there exists some basis whose $\mathbb{Z}$-linear combination is preserved by every element of W. Hence, the matrices corresponding to this basis contain integer entries. Thus, the trace will be an integer. Since similar matrices have the same trace, it follows that the matrix corresponding to every element of W (with respect to any basis) has integral trace.

Proposition 2.2.2. *If W is a crystallographic group and $\alpha \neq \beta$, then the possible values of $m(\alpha, \beta)$ are $2, 3, 4, 6$.*

Proof. If $\alpha \neq \beta$, then $s_\alpha \neq s_\beta$, and hence $s_\alpha s_\beta \neq 1$. Since $s_\alpha s_\beta$ is a rotation by an angle $\frac{2\pi}{m(\alpha,\beta)}$, it fixes the orthogonal complement of the spanning set of α, β. Thus the trace of the matrix corresponding to $s_\alpha s_\beta$ is $n-2+2\cos\theta$. The trace is an integer, and thus $\cos\theta \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$, but not 1, as the angle is not zero. So $\theta \in \{\pi, \frac{2\pi}{3}, \frac{\pi}{2}, \frac{\pi}{3}\}$, which implies $m(\alpha, \beta) \in \{2, 3, 4, 6\}$. □

Definition 2.2.3 (Crystallographic Root System)**.** *A root system Φ is said to be crystallographic if it satisfies one additional condition, namely that $2\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle} \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$.*

Definition 2.2.4 (Weyl Group)**.** *The group generated by a crystallographic root system is known as a Weyl group.*

1. It is enough to check only for $\alpha, \beta \in \Delta$. This is because of the following reason. Let $2\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle} \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$. Now, we know that there exists

$w \in W$ such that $w\alpha \in \Delta$, as $W\Delta = \Phi$. Since each w is an orthogonal transformation, we have $2\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle} = 2\frac{\langle w\alpha,w\beta\rangle}{\langle w\beta,w\beta\rangle}$, where $w\beta \in \Delta$ and $w\alpha \in \Phi$. Now, we write $w\alpha$ as a $\mathbb{Z}$-linear combination of the members of Δ. Then $2\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle} = 2\frac{\langle w\alpha,w\beta\rangle}{\langle w\beta,w\beta\rangle} = 2\frac{\langle w\alpha,\sum c_\gamma\gamma\rangle}{\langle w\beta,w\beta\rangle} = 2\sum c_\gamma \frac{\langle w\alpha,\gamma\rangle}{\langle w\alpha,w\alpha\rangle}$, so if the ratio in each term belongs to $\mathbb{Z}$, it follows that $2\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle} \in \mathbb{Z}$.

2. It is clear that all roots are in the $\mathbb{Z}$-linear combination of Δ.

3. We conclude that every Weyl group is crystallographic, i.e., every Weyl group preserves a $\mathbb{Z}$-linear combination of some basis (in particular, the $\mathbb{Z}$-linear combination of Δ in this case). Let γ be in the $\mathbb{Z}$-linear combination of Δ. Then $\gamma = \sum c_\beta\beta$ for all $\beta \in \Delta$. Taking any $w \in W$, we have $w\gamma = \sum c_\beta(w\beta)$, where $c_\beta \in \mathbb{Z}$ and $w(\beta) \in \Phi$, which implies that $w\gamma$ is in the $\mathbb{Z}$-linear combination of Φ and hence of Δ.

4. The converse is also true: every crystallographic group is a Weyl group.

5. We can have different crystallographic root systems which correspond to the same Weyl group. This happens only in the case of B_n and not in other cases. We can see this using connected Coxeter graphs.
   There exist 2 root systems B_n and C_n corresponding to the Weyl group B_n, which are non-isomorphic. They are non-isomorphic because an isomorphism of root systems Φ_1 and Φ_2 is defined as a linear map preserving the inner product from $V_1 \to V_2$. Since B_n and C_n have different numbers of long and short roots, they are not isomorphic.

6. Now, note that from the graphs listed above which are positive definite, we can conclude that when W is irreducible, meaning Γ is connected, at most 2 root lengths are possible. This is because we only need to examine the lengths of simple roots, as $W\Delta = \Phi$. When 2 simple roots are connected by an unlabelled edge, they form $I_2(3)$. In $I_2(3)$, all reflections are W-conjugate and hence have the same length. So, in the case when the highest label in the Coxeter diagram is 3, all simple roots and hence all roots have the same length. This is because $ws_\alpha w^{-1} = s_\beta \implies s_{w(\alpha)} = s_\beta \implies w(\alpha) = \pm\beta \implies |\alpha| = |\beta|$. Note that in the diagrams of connected Coxeter graphs, at most 1 label is mentioned, and hence we can conclude that if Φ is irreducible, then it can have at most 2 lengths possible.

7. Now, we can also determine the possible ratios of lengths of roots in the case when 2 lengths are possible. Let α and β be 2 roots of Δ. Consider $m = 2\frac{\langle\alpha,\beta\rangle}{\langle\alpha,\alpha\rangle}$ and $n = 2\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle}$. Then $m, n \in \mathbb{Z}$, which implies $4\frac{\langle\alpha,\beta\rangle^2}{\langle\alpha,\alpha\rangle\langle\beta,\beta\rangle} \in$

$\mathbb{Z}$. By the Cauchy–Schwarz inequality, $mn < 4$, and since $mn \in \mathbb{Z}$, it follows that $m, n \in \{1, 2, 3\}$. It follows that $\frac{m}{n} \in \{1, 2, 3, \frac{1}{2}, \frac{1}{3}\}$. Hence the possible ratios of lengths of α and β are $\{1, \sqrt{2}, \sqrt{3}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}\}$.

Also note that we know the ratio between α and β when $m(\alpha, \beta)$ is known. We want to find the ratio when $m(\alpha, \beta)$ is either 3, 4, or 6. The case $m(\alpha, \beta) = 3$ has been discussed above, where it is known that both roots must have the same length. However, we now compute the ratio using a new approach for $m(\alpha, \beta) = 3$ as well.
Note that from above, $mn = 4\frac{\langle\alpha,\beta\rangle^2}{\langle\alpha,\alpha\rangle\langle\beta,\beta\rangle} = 4\cos^2\left(\frac{\theta}{2}\right)$, where $\frac{\theta}{2}$ is the angle between α and β, and θ is the angle of rotation of $s_\alpha s_\beta$ from the previous proposition.
When $m(\alpha, \beta) = 3$, then $\theta = \frac{2\pi}{3}$, which implies $\frac{\theta}{2} = \frac{\pi}{3}$, and hence $mn = 1$, so $\frac{m}{n} = 1$, meaning α and β have the same length.
When $m(\alpha, \beta) = 4$, then $\theta = \frac{2\pi}{4}$, which implies $\frac{\theta}{2} = \frac{\pi}{4}$, and hence $mn = 2$, so $\frac{m}{n} = 2$ or $\frac{1}{2}$, meaning the ratio of squared lengths of α and β is 2 or $\frac{1}{2}$.
When $m(\alpha, \beta) = 6$, then $\theta = \frac{2\pi}{6}$, which implies $\frac{\theta}{2} = \frac{\pi}{3}$, and hence $mn = 3$, so $\frac{m}{n} = 3$ or $\frac{1}{3}$, meaning the ratio of squared lengths of α and β is 3 or $\frac{1}{3}$.

2.3 Construction of Root Systems and the highest root.

We will now construct all root systems for each of the Weyl groups. Our strategy is as follows: we first choose a lattice in V, and then choose vectors of particular lengths, preferably only two, as we know that the root system of a connected Weyl group can have at most two lengths. Also, we know from the Coxeter graphs which groups have two lengths and which have only one. We then choose accordingly and check the condition

$$\frac{2\langle\alpha, \beta\rangle}{\langle\alpha, \alpha\rangle} \in \mathbb{Z}$$

for all $\alpha, \beta \in \Phi$. Actually, it is enough to check this for all members of Δ only. This is because if we assume that for all $\alpha, \beta \in \Delta$, $\frac{2\langle\alpha,\beta\rangle}{\langle\alpha,\alpha\rangle} \in \mathbb{Z}$, then we can prove it for all members of Φ as well. Let $\alpha, \beta \in \Phi$. We know that $W\Delta = \Phi$, hence there exists $w \in W$ such that $w\alpha \in \Delta$. Now,

$$\frac{2\langle\alpha, \beta\rangle}{\langle\alpha, \alpha\rangle} = \frac{2\langle w\alpha, w\beta\rangle}{\langle w\alpha, w\alpha\rangle}.$$

We write $w\beta$ as a $\mathbb{Z}$-linear combination of members of Δ:

$$w\beta = \sum_{\gamma\in\Delta} c_\gamma \gamma.$$

Thus

$$\frac{2\langle w\alpha, w\beta\rangle}{\langle w\alpha, w\alpha\rangle} = \frac{2\left\langle w\alpha, \sum_{\gamma\in\Delta} c_\gamma\gamma\right\rangle}{\langle w\alpha, w\alpha\rangle} = \sum_{\gamma\in\Delta} c_\gamma \left(\frac{2\langle w\alpha, \gamma\rangle}{\langle w\alpha, w\alpha\rangle}\right).$$

Since each factor belongs to $\mathbb{Z}$, the whole sum is in $\mathbb{Z}$.

Now we claim that checking the above condition is enough to conclude that Φ is a root system. Since any lattice contains only finitely many vectors of particular lengths, Φ is finite. Next, we verify that $s_\alpha(\Phi) = \Phi$ for all $\alpha \in \Phi$. Take any $\beta \in \Phi$; then

$$s_\alpha(\beta) = \beta - \frac{2\langle\alpha, \beta\rangle}{\langle\alpha, \alpha\rangle}\alpha.$$

Note that $s_\alpha(\beta)$ is a $\mathbb{Z}$-linear combination of α and β, which implies $s_\alpha(\beta) \in L$. Also, $s_\alpha(\beta)$ and β have the same length, therefore $s_\alpha(\beta) \in \Phi$. Now we want to show that $L_\alpha \cap \Phi = \{\alpha, -\alpha\}$. This is because if $\alpha \in \Phi$, then the only scalar multiples of α which can be in L are $-\alpha$, $\pm\sqrt{2}\,\alpha$, $\pm\sqrt{3}\,\alpha$, $\pm\frac{1}{\sqrt{2}}\alpha$, and $\pm\frac{1}{\sqrt{3}}\alpha$, but none of these except $-\alpha$ lie in the lattice. Thus $L_\alpha \cap \Phi = \{\alpha, -\alpha\}$.

Root System of A_n :
Take Take $V = \{x = (x_1, x_2, \cdots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1 + x_2 + \cdots + x_{n+1} = 0\}$. Take L' to be the standard lattice and choose all vectors of length $\sqrt{2}$. We get

$$\Phi = \left\{\ e_i - e_j \quad (i \neq j,\ i, j \in \{1, \cdots, n+1\},\ \text{giving } n(n+1))\ \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = e_1 - e_2, \\ \alpha_2 = e_2 - e_3, \\ \vdots \\ \alpha_n = e_n - e_{n+1} \end{array} \right\}$$

$$\Pi = \left\{\ e_i - e_j \quad (i < j,\ i, j \in \{1, \cdots, n+1\})\ \right\}$$

So, the highest root is $e_1 - e_{n+1}$.

Root System of B_n :
Take $V = \mathbb{R}^n$. Take L' to be the standard lattice and choose all vectors of

squared length 1 or 2. We get

$$\Phi = \left\{ \begin{array}{ll} \pm e_i & (i \in \{1, \cdots, n\}) \\ \pm e_i \pm e_j & (i < j) \end{array} \right\}$$

which expands to

$$\Phi = \left\{ \begin{array}{ll} e_i - e_j & (i < j, \text{ giving } \frac{n(n-1)}{2}), \\ -e_i + e_j & (i < j, \text{ giving } \frac{n(n-1)}{2}), \\ e_i + e_j & (i < j, \text{ giving } \frac{n(n-1)}{2}), \\ -e_i - e_j & (i < j, \text{ giving } \frac{n(n-1)}{2}), \\ \pm e_i & (\text{giving } 2n) \end{array} \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = e_1 - e_2, \\ \alpha_2 = e_2 - e_3, \\ \vdots \\ \alpha_{n-1} = e_{n-1} - e_n, \\ \alpha_n = e_n \end{array} \right\}$$

Now note that,

$$\begin{array}{l} e_n = \alpha_n, \\ e_{n-1} = \alpha_{n-1} + \alpha_n, \\ \vdots \\ e_1 = \alpha_1 + \alpha_2 + \cdots + \alpha_n \end{array}$$

So clearly the members of Π are

$$\Pi = \left\{ \begin{array}{ll} e_i - e_j & (i < j), \\ e_i + e_j & (i < j), \\ e_i & (i \in \{1, \cdots, n\}) \end{array} \right\}$$

Among these, the highest root clearly comes from the set $\{e_i + e_j,\, i < j\}$, which is $e_1 + e_2$.

Root System of C_n:

Take $V = \mathbb{R}^n$. Take L' to be the standard lattice and choose all vectors of squared length 2 or 4. We get

$$\Phi = \left\{ \begin{array}{ll} \pm 2e_i & (i \in \{1, \cdots, n\}) \\ \pm e_i \pm e_j & (i < j) \end{array} \right\}$$

which expands to

$$\Phi = \left\{ \begin{array}{ll} e_i - e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ -e_i + e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ e_i + e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ -e_i - e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ \pm 2e_i & (\text{giving } 2n) \end{array} \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = e_1 - e_2, \\ \alpha_2 = e_2 - e_3, \\ \vdots \\ \alpha_{n-1} = e_{n-1} - e_n, \\ \alpha_n = 2e_n \end{array} \right\}$$

$$\begin{array}{l} e_n = \frac{1}{2}\alpha_n, \\ e_{n-1} = \alpha_{n-1} + \frac{1}{2}\alpha_n, \\ \vdots \\ e_1 = \alpha_1 + \alpha_2 + \cdots + \alpha_{n-1} + \frac{1}{2}\alpha_n \end{array}$$

$$\Pi = \left\{ \begin{array}{ll} e_i - e_j & (i<j), \\ e_i + e_j & (i<j), \\ 2e_i & (i \in \{1, \cdots, n\}) \end{array} \right\}$$

Among these, the highest root clearly comes from the set $\{2e_i\}$, which is $2e_1$.

Root System of D_n:

Take $V = \mathbb{R}^n$. Take L' to be the standard lattice and choose all vectors of squared length 2. We get

$$\Phi = \left\{ \ \pm e_i \pm e_j \quad (i<j) \ \right\}$$

which expands to

$$\Phi = \left\{ \begin{array}{ll} e_i - e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ -e_i + e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ e_i + e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}), \\ -e_i - e_j & (i<j, \text{ giving } \frac{n(n-1)}{2}) \end{array} \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = e_1 - e_2, \\ \alpha_2 = e_2 - e_3, \\ \vdots \\ \alpha_{n-1} = e_{n-1} - e_n, \\ \alpha_n = e_{n-1} + e_n \end{array} \right\}$$

$$
\begin{aligned}
&e_n = \tfrac{1}{2}(\alpha_n - \alpha_{n-1}),\\
&e_{n-1} = \tfrac{1}{2}(\alpha_{n-1} + \alpha_n),\\
&e_{n-2} = \alpha_{n-2} + \tfrac{1}{2}(\alpha_{n-1} + \alpha_n),\\
&\vdots\\
&e_1 = \alpha_1 + \alpha_2 + \cdots + \alpha_{n-2} + \tfrac{1}{2}(\alpha_{n-1} + \alpha_n)
\end{aligned}
$$

$$
\Pi = \left\{ \begin{array}{ll} e_i - e_j & (i<j),\\ e_i + e_j & (i<j) \end{array} \right\}
$$

Among these, the highest root clearly comes from the set $\{e_i + e_j,\, i<j\}$, which is $e_1 + e_2$.

Root System of G_2:

Let $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$. Define Φ to be the set of vectors of squared length 2 or 6 in the intersection of V with the standard lattice. So Φ consists of six short roots $\pm(e_i - e_j)$ $(i<j)$ and six long roots $\pm(2e_i - e_j - e_k)$ where $\{i,j,k\} = \{1,2,3\}$.

$$
\Phi = \left\{ \begin{array}{ll} \pm(e_i - e_j) & (i<j,\ \text{giving 6}),\\ \pm(2e_i - e_j - e_k) & (\{i,j,k\} = \{1,2,3\},\ \text{giving 6}) \end{array} \right\}
$$

$$
\Delta = \left\{ \begin{array}{l} \alpha_1 = e_1 - e_2,\\ \alpha_2 = -2e_1 + e_2 + e_3 \end{array} \right\}
$$

Among these, the highest root is $\tilde{\alpha} = 2e_3 - e_1 - e_2$.

Root System of F_4 :

Choose the lattice $L = L' + \mathbb{Z}\left(\frac{1}{2}\sum_{i=1}^{4} e_i\right)$, where L' is the standard lattice. We know that the Coxeter diagram of F_4 contains a unique label 4, and hence there exist vectors of two possible lengths having ratio $\sqrt{2}$. So choose vectors of length 1 and $\sqrt{2}$ in this lattice. We get

$$
\Phi = \left\{ \begin{array}{ll} \pm e_i \pm e_j & (i<j),\\ \pm e_i, & \\ \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4) & \end{array} \right\}
$$

implies that

$$
\Phi = \left\{ \begin{array}{ll}
e_i + e_j & (i<j,\ \text{giving 6}),\\
-e_i - e_j & (i<j,\ \text{giving 6}),\\
-e_i + e_j & (i<j,\ \text{giving 6}),\\
e_i - e_j & (i<j,\ \text{giving 6}),\\
e_i & (4),\\
-e_i & (4),\\
\frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4) & (16)
\end{array} \right\}
$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = e_2 - e_3, \\ \alpha_2 = e_3 - e_4, \\ \alpha_3 = e_4, \\ \alpha_4 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4) \end{array} \right\}$$

Now we intend to calculate the highest root. First note that

$$\begin{aligned} e_4 &= \alpha_3, \\ e_3 &= \alpha_3 + \alpha_2, \\ e_2 &= \alpha_3 + \alpha_2 + \alpha_1, \\ e_1 &= 2\alpha_4 + 3\alpha_3 + 2\alpha_2 + \alpha_1. \end{aligned}$$

So clearly the members of Π are

$$\Pi = \left\{ \begin{array}{ll} e_i + e_j & (i < j), \\ e_i - e_j & (i < j), \\ e_i, & \\ \frac{1}{2}(e_1 \pm e_2 \pm e_3 \pm e_4) & \end{array} \right\}$$

Among these, the highest root clearly comes from the set $\{e_i + e_j, i < j\}$, which is $e_1 + e_2$.

Root System of E_8 :
Take $V = \mathbb{R}^8$. Now take L' consisting of all $\sum c_i e_i$ with $c_i \in \mathbb{Z}$ and $\sum c_i$ even. Then $L = L' + \mathbb{Z}\left(\frac{1}{2}\sum_{i=1}^{8} e_i\right)$. Now, recall the Coxeter graph of E_8, there is no labelling, hence we take only single length roots from L. Here $|\Phi| = 240$.

$$\Phi = \left\{ \begin{array}{ll} e_i + e_j & (i < j, \text{ giving } 28), \\ -e_i - e_j & (i < j, \text{ giving } 28), \\ e_i - e_j & (i < j, \text{ giving } 28), \\ -e_i + e_j & (i < j, \text{ giving } 28), \\ \frac{1}{2}\sum_{i=1}^{8}(\pm e_i) & (\text{with even number of positive signs, giving } 128) \end{array} \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 - e_8), \\ \alpha_2 = e_1 + e_2, \\ \alpha_3 = e_2 - e_1, \\ \alpha_4 = e_3 - e_2, \\ \alpha_5 = e_4 - e_3, \\ \alpha_6 = e_5 - e_4, \\ \alpha_7 = e_6 - e_5, \\ \alpha_8 = e_7 - e_6 \end{array} \right\}$$

Now note that

$$
\begin{aligned}
e_1 &= \tfrac{1}{2}(\alpha_2 - \alpha_3),\\
e_2 &= \tfrac{1}{2}(\alpha_2 + \alpha_3),\\
e_3 &= \tfrac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4,\\
e_4 &= \tfrac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5,\\
e_5 &= \tfrac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6,\\
e_6 &= \tfrac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6 + \alpha_7,\\
e_7 &= \tfrac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6 + \alpha_7 + \alpha_8,\\
e_8 &= 2\alpha_1 + \tfrac{5}{2}\alpha_2 + \tfrac{7}{2}\alpha_3 + 5\alpha_4 + 4\alpha_5 + 3\alpha_6 + 2\alpha_7 + \alpha_8
\end{aligned}
$$

$$
\Pi = \left\{ \begin{array}{ll}
e_i + e_j & (i < j),\\
-e_i + e_j & (i < j),\\
\frac{1}{2}\left(\sum_{i=1}^{7}(\pm e_i) + e_8\right) & \text{(with even number of positive signs)}
\end{array} \right\}
$$

Among these, the highest root clearly comes from the set $\{e_i + e_j,\, i < j\}$, which is $e_7 + e_8$.

Root System of E_7 :

$$\Phi = \left\{ \begin{array}{ll} e_i + e_j & (i < j,\, i,j \in \{1,\cdots,6\},\ \text{giving 15}), \\ -e_i - e_j & (i < j,\, i,j \in \{1,\cdots,6\},\ \text{giving 15}), \\ -e_i + e_j & (i < j,\, i,j \in \{1,\cdots,6\},\ \text{giving 15}), \\ e_i - e_j & (i < j,\, i,j \in \{1,\cdots,6\},\ \text{giving 15}), \\ \pm(e_7 - e_8) & (\text{giving 2}), \\ \frac{1}{2}\left(e_7 - e_8 + \sum_{i=1}^{6}(\pm e_i)\right) & (\text{odd number of minus signs in sum, giving 32}), \\ \frac{-1}{2}\left(e_7 + e_8 + \sum_{i=1}^{6}(\pm e_i)\right) & (\text{odd number of minus signs in sum, giving 32}) \end{array} \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 - e_8), \\ \alpha_2 = e_1 + e_2, \\ \alpha_3 = e_2 - e_1, \\ \alpha_4 = e_3 - e_2, \\ \alpha_5 = e_4 - e_3, \\ \alpha_6 = e_5 - e_4, \\ \alpha_7 = e_6 - e_5 \end{array} \right\}$$

Now note that,

$$\begin{array}{l} e_1 = \frac{1}{2}(\alpha_2 - \alpha_3), \\ e_2 = \frac{1}{2}(\alpha_2 + \alpha_3), \\ e_3 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4, \\ e_4 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5, \\ e_5 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6, \\ e_6 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6 + \alpha_7, \\ e_7 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6 + \alpha_7 + \alpha_8, \\ e_8 - e_7 = 2\alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + \alpha_7 \end{array}$$

$$\Pi = \left\{ \begin{array}{ll} e_i + e_j & (i < j,\, i,j \in \{1,\cdots,6\}), \\ -e_i + e_j & (i < j,\, i,j \in \{1,\cdots,6\}), \\ e_8 - e_7, & \\ \frac{-1}{2}\left(e_7 - e_8 + \sum_{i=1}^{6}(\pm e_i)\right) & (\text{odd number of minus signs in sum}) \end{array} \right\}$$

Among these the highest root is $e_8 - e_7$.

Root System of E_6 :

$$\Phi = \left\{ \begin{array}{ll} e_i + e_j & (i<j,\, i,j \in \{1,\cdots,5\},\ \text{giving 10}), \\ -e_i - e_j & (i<j,\, i,j \in \{1,\cdots,5\},\ \text{giving 10}), \\ -e_i + e_j & (i<j,\, i,j \in \{1,\cdots,5\},\ \text{giving 10}), \\ e_i - e_j & (i<j,\, i,j \in \{1,\cdots,5\},\ \text{giving 10}), \\ \frac{1}{2}\left(e_8 - e_7 - e_6 + \sum_{i=1}^{5}(\pm e_i)\right) & (\text{even number of minus signs in sum, giving 16}), \\ \frac{-1}{2}\left(e_8 - e_7 - e_6 + \sum_{i=1}^{5}(\pm e_i)\right) & (\text{even number of minus signs in sum, giving 16}) \end{array} \right\}$$

$$\Delta = \left\{ \begin{array}{l} \alpha_1 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 - e_8), \\ \alpha_2 = e_1 + e_2, \\ \alpha_3 = e_2 - e_1, \\ \alpha_4 = e_3 - e_2, \\ \alpha_5 = e_4 - e_3, \\ \alpha_6 = e_5 - e_4 \end{array} \right\}$$

Now note that,

$$\begin{array}{l} e_1 = \frac{1}{2}(\alpha_2 - \alpha_3), \\ e_2 = \frac{1}{2}(\alpha_2 + \alpha_3), \\ e_3 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4, \\ e_4 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5, \\ e_5 = \frac{1}{2}(\alpha_2 + \alpha_3) + \alpha_4 + \alpha_5 + \alpha_6, \\ e_8 - e_7 - e_6 = 2\alpha_1 + \frac{3}{2}\alpha_2 + \frac{5}{2}\alpha_3 + 3\alpha_4 + 2\alpha_5 + \alpha_6 \end{array}$$

$$\Pi = \left\{ \begin{array}{ll} e_i + e_j & (i<j,\, i,j \in \{1,\cdots,5\}), \\ -e_i + e_j & (i<j,\, i,j \in \{1,\cdots,5\}), \\ \frac{1}{2}\left(e_8 - e_7 - e_6 + \sum_{i=1}^{5}(\pm e_i)\right) & (\text{even number of minus signs in sum}) \end{array} \right\}$$

So, among these the highest root is $\frac{1}{2}(e_8 - e_7 - e_6 + e_5 + e_4 + e_3 + e_2 + e_1)$.

Realization as Abstract Groups

The finite reflection groups in the infinite families admit concrete realizations as groups of linear transformations.

- The group $W(A_n)$ acts on the vector space $V = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1 + \cdots + x_{n+1} = 0\}$ by permuting coordinates. Under this action, $W(A_n) \cong S_{n+1}$.

- The group $W(B_n)$ acts on $\mathbb{R}^n$ by permuting coordinates and changing their signs independently. Thus $W(B_n) \cong C_2^n \rtimes S_n$.

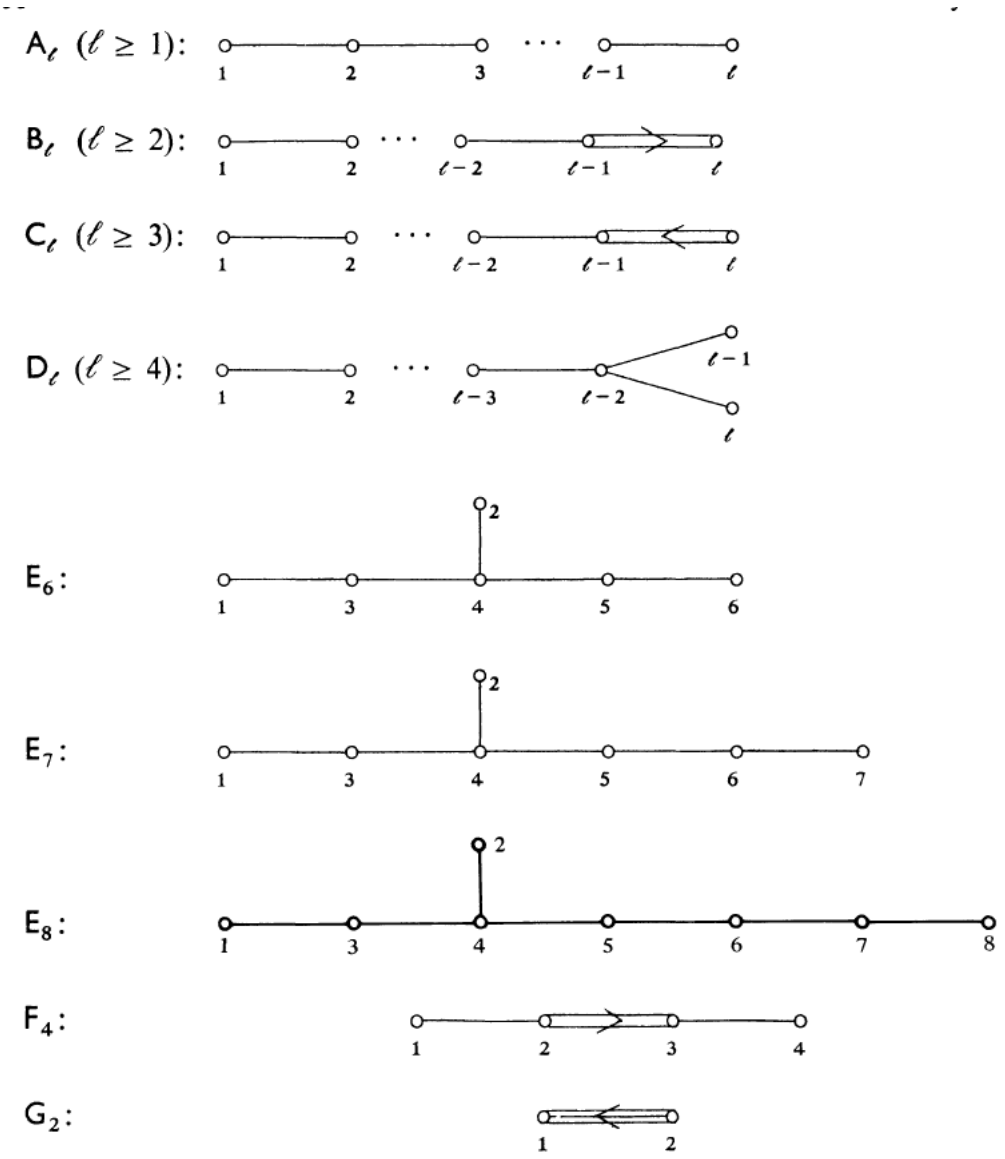
- The group $W(D_n)$ acts on $\mathbb{R}^n$ by permuting coordinates and changing the signs of an even number of coordinates. Hence $W(D_n) \cong C_2^{n-1} \rtimes S_n$.

- The group $W(I_2(m))$ acts on $\mathbb{R}^2$ as the symmetry group of a regular m-gon. Consequently, $W(I_2(m)) \cong D_{2m}$.

These realizations provide geometric interpretations of the corresponding abstract Coxeter groups.For the exceptional types E_6, E_7, E_8, F_4, H_3, and H_4, there are no realizations analogous to those of the infinite families in terms of familiar groups such as symmetric or dihedral groups. Consequently, their structure is less transparent from the abstract group-theoretic point of view. To obtain further information about these exceptional reflection groups, we shall compute their orders directly from the corresponding root systems and Dynkin diagrams.

2.4 Dynkin Diagrams, and Computing the Order of W

Definition 2.4.1 (Dynkin Diagram)**.** *From the connected Coxeter graph, replace the label* 4 *and* 6 *by a double and triple edge respectively, and direct an arrow from the long root to the short root.*



For the classical families A_n, B_n, D_n, and the dihedral groups $I_2(m)$, the order of the corresponding reflection groups can be determined directly from their natural realizations. However, for the exceptional types, the order is not immediately obvious even when the root system is explicitly known. A general method for computing $|W|$ is obtained from the action of W on its root system Φ.

Recall that if a finite group G acts on a set X, then for any $x \in X$, the orbit–stabilizer theorem gives

$$|G| = |Gx| \cdot |G_x|,$$

where Gx denotes the orbit of x and G_x is the stabilizer of x. Applying this to the action of a Weyl group W on the set of roots Φ, one can compute the order of W by determining the size of the orbit of a suitable root and the order of its stabilizer.

For an irreducible root system, all roots of the same length lie in a single W-orbit. In particular, one may choose the highest root $\widetilde{\alpha}$. The stabilizer of $\widetilde{\alpha}$ is generated by the reflections corresponding to those simple roots that are orthogonal to $\widetilde{\alpha}$. Consequently, the type of the stabilizer can be read directly from the Dynkin diagram, which leads to an inductive computation of the order of W.

- **Type F_4.** There are 24 long roots. The highest root $\widetilde{\alpha}$ is orthogonal to all simple roots except α_1, and hence its stabilizer is of type C_3, whose order is 48. Therefore,

$$|W(F_4)| = 24 \times 48 = 1152 = 2^7 \cdot 3^2.$$

- **Type E_6.** All 72 roots form a single orbit. The highest root is orthogonal to all simple roots except α_2, so the stabilizer is of type A_5, whose order is 6!. Hence

$$|W(E_6)| = 72 \cdot 6! = 2^7 \cdot 3^4 \cdot 5.$$

- **Type E_7.** The root system contains 126 roots forming one orbit. The highest root is orthogonal to all simple roots except α_1, and therefore its stabilizer is of type D_6, whose order is $2^5 \cdot 6!$. Thus

$$|W(E_7)| = 126 \cdot 2^5 \cdot 6! = 2^{10} \cdot 3^4 \cdot 5 \cdot 7.$$

- **Type E_8.** The root system has 240 roots. The highest root is orthogonal to all simple roots except α_8, and hence its stabilizer is of type E_7. Using the order of $W(E_7)$ obtained above, we find

$$|W(E_8)| = 240\,|W(E_7)| = 2^{14} \cdot 3^5 \cdot 5^2 \cdot 7.$$

The orders of all irreducible Weyl groups together with the number of roots are summarized in Table 2.1. Thus, the orbit–stabilizer theorem provides a uniform and inductive approach to computing the orders of exceptional Weyl groups, complementing the direct computations available for the classical families.

Table 2.1: Orders of irreducible Weyl groups and the number of roots

Type	$\lvert W\rvert$	$\lvert\Phi\rvert$
A_n	$(n+1)!$	$n(n+1)$
$B_n = C_n$	$2^n n!$	$2n^2$
D_n	$2^{n-1} n!$	$2n(n-1)$
E_6	$2^7 \cdot 3^4 \cdot 5$	72
E_7	$2^{10} \cdot 3^4 \cdot 5 \cdot 7$	126
E_8	$2^{14} \cdot 3^5 \cdot 5^2 \cdot 7$	240
F_4	$2^7 \cdot 3^2$	48
G_2	12	12

Chapter 3

Polynomial Invariants of Finite Reflection Groups

In this chapter, we study the invariant theory of finite reflection groups. While the previous chapters focused on the geometric and combinatorial structure of reflection groups through root systems, Coxeter graphs, and Weyl groups, our objective here is to extract algebraic information about these groups from their actions on polynomial rings. Given a finite reflection group W acting on a vector space V, the induced action on the polynomial algebra $S(V^*)$ allows us to consider the collection of polynomials that remain unchanged under the action of W. These polynomials form the invariant ring, denoted by $S(V^*)^W$.

A remarkable feature of finite reflection groups is that their invariant ring possesses a particularly simple structure. A fundamental theorem of Chevalley, Shephard, and Todd states that the invariant ring is a polynomial algebra generated by algebraically independent homogeneous polynomials, called the *basic invariant polynomials*. Thus, much of the information about the group is encoded in the degrees of these basic invariants.

The main goal of this chapter is to investigate the relationship between the structure of a finite reflection group and the degrees of its basic invariant polynomials. We shall see that these degrees are far from arbitrary and satisfy several remarkable numerical relations. In particular, the product of the degrees of the basic invariants equals the order of the reflection group W, while the sum of these degrees is equal to $N+n$, where N denotes the number of reflections in W and n is the number of basic invariants. These results reveal a deep connection between the algebraic structure of the invariant ring and the geometric structure of the reflection group.

The chapter culminates in the study of Coxeter elements and exponents, leading to the fundamental relation between the exponents of a finite reflec-

tion group and the degrees of its basic invariant polynomials. This establishes one of the most important links between invariant theory, geometry, and the theory of finite reflection groups.

3.1 Generating set of Polynomial Invariants of Finite Reflection Groups

Let G be a finite group acting on a Euclidean space V. There is a natural action of G on V^*, the dual space, defined by $f(g(v)) = g^{-1}(f(v))$. Let $\{x_1, x_2, \cdots, x_n\}$ be the coordinate functions of V^*, and define $S(V^*)$ to be the set of all polynomials in the variables $\{x_1, x_2, \cdots, x_n\}$. Thus, there is an action of G on S.

Definition 3.1.1 (Polynomial Invariants of a Finite Group)**.** *A polynomial f such that $g \cdot f = f$ for all $g \in G$ is called a polynomial invariant.*

The set of all such invariant polynomials forms a subring of S, denoted by R. As a K-algebra, R is finitely generated and has transcendence degree n, where n is the dimension of V.
Define L to be the field of fractions of S, and L^G to be the invariant elements of L under G. Then L^G is equal to the field of fractions of R. It follows that R, the set of all invariant polynomials, has transcendence degree n, where n is the dimension of the vector space V.

Definition 3.1.2 (Averaging Operator)**.** *Define*

$$f^{\#} = \frac{1}{|G|} \sum_{g \in G} g \cdot f$$

, $f^{\#}$ is known as the averaging operator of f.

We know that any ideal of a polynomial ring is finitely generated (Hilbert's Basis Theorem). However, R, the set of all invariants, is not an ideal of S. So, define R^+ to be the set of all positive degree invariant polynomials, and let $I = SR^+$ be the ideal generated by R^+. By Hilbert's Basis Theorem, this ideal is finitely generated.

Proposition 3.1.3. *Let I be the ideal generated by R^+, and let the generators of I be $\{f_1, f_2, \cdots, f_m\}$, which are homogeneous and all of positive degree. Then, as an algebra, R is generated by these elements together with 1.*

Lemma 3.1.4. *Let $f_1, f_2, \cdots, f_r \in R$, where f_1 is not in the ideal of R generated by $f_2, \cdots, f_r$. If $g_1, g_2, \cdots, g_r$ are homogeneous elements of S such that $f_1 g_1 + f_2 g_2 + \cdots + f_r g_r = 0$, then $g_1 \in I$.*

Theorem 3.1.5 (**Chevalley's Theorem**). *Let W be a finite reflection group and S be the set of all polynomials in the variables $\{x_1, x_2, \cdots, x_n\}$, which are the coordinate functions of V^*. If R is the set of all invariant polynomials, then R is generated by n homogeneous algebraically independent elements of positive degree.*

3.2 Degrees of Basic polynomial invariants

Proposition 3.2.1 (Uniqueness of Degrees). *Let $\{f_1, f_2, \cdots, f_n\}$ and $\{g_1, g_2, \cdots, g_n\}$ be two different sets of n homogeneous algebraically independent generators of R with degrees d_i and e_i respectively. Then, after renumbering one of the sets, we get $d_i = e_i$, that is, the multiset of degrees of both sets is the same.*

Proof. Since the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically independent, we have $\frac{\partial f_i}{\partial f_j} = \delta_{ij}$, where δ_{ij} is the Kronecker delta. Also, since each f_i can be written in terms of $g_1, g_2, \cdots, g_n$, we can compute $\frac{\partial f_i}{\partial f_j}$ using the chain rule in terms of g_j:

$$\frac{\partial f_i}{\partial f_j} = \sum_{k=1}^{n} \frac{\partial f_i}{\partial g_k} \frac{\partial g_k}{\partial f_j}.$$

Combining the above two equations gives

$$\frac{\partial f_i}{\partial f_j} = \sum_{k=1}^{n} \frac{\partial f_i}{\partial g_k} \frac{\partial g_k}{\partial f_j} = \delta_{ij}.$$

Viewing the above equation in terms of matrices, let $A = \left(\frac{\partial f_i}{\partial g_k}\right)$ and $B = \left(\frac{\partial g_k}{\partial f_j}\right)$. Then it is clear that $AB = I$, which implies that $\det(A) \neq 0$. By the definition of the determinant,

$$\det(A) = \sum_{\pi} (\pm) \prod_{i=1}^{n} \frac{\partial f_i}{\partial g_{\pi(i)}}.$$

Since the determinant is non-zero, there exists a permutation π for which $\prod_{i=1}^{n} \frac{\partial f_i}{\partial g_{\pi(i)}} \neq 0$. After renumbering the set $\{g_1, g_2, \cdots, g_n\}$, we can arrange that $\prod_{i=1}^{n} \frac{\partial f_i}{\partial g_i} \neq 0$. It follows that whenever f_i is written in terms of $g_1, g_2, \cdots, g_n$, the element g_i must occur. Hence $d_i \geq e_i$. Changing the roles of $\{f_1, f_2, \cdots, f_n\}$ and $\{g_1, g_2, \cdots, g_n\}$, we get $e_i \geq d_i$.Hence $d_i = e_i$, which means the multiset of degrees of both sets is the same. □

Next, we intend to show the order of W in terms of the degrees of the generators of R. For that, we need a proposition, and to prove that proposition, we intend to prove a lemma.

Lemma 3.2.2. *Let E be a vector space of dimension n over a field of characteristic zero, on which W, a finite reflection group, acts. Let E^W denote the set of all invariant elements of E. Then the dimension of E^W is equal to the trace of the averaging operator.*

Proof. Let z be the averaging linear operator; mathematically, $z = \frac{1}{|W|}\sum_{w\in W} w$. We intend to prove that the trace of z is equal to the dimension of E^W. The approach is to prove that z is idempotent and hence diagonalisable with eigenvalues 0 or 1 only. Denote by E^1 the eigenspace of z corresponding to eigenvalue 1. We then prove that $E^1 = E^W$ and hence conclude our requirement.

Note that $w \cdot z = z$ for all $w \in W$. This is because

$$w \cdot z = w \cdot \frac{1}{|W|}\sum_{w'\in W} w' = \frac{1}{|W|}\sum_{w'\in W}(ww') = \frac{1}{|W|}\sum_{w'\in W} w' = z.$$

Next, we intend to prove that z is idempotent. This follows since

$$z^2 = \left(\frac{1}{|W|}\sum_{w\in W} w\right) z = \frac{1}{|W|}\sum_{w\in W} wz = \frac{1}{|W|}\sum_{w\in W} z = \frac{1}{|W|}|W|z = z.$$

Hence, z is idempotent and therefore diagonalisable, so it can have eigenvalues 0 or 1 only. Thus, the multiplicity of 1 is equal to the trace of the linear operator z. Let E^1 be the eigenspace corresponding to eigenvalue 1.

Next, we intend to prove that $E^W = E^1$. Let $e \in E^W$, which implies that $we = e$ for all $w \in W$. We want to show that $z \cdot e = e$:

$$z \cdot e = \frac{1}{|W|}\left(\sum_{w\in W} w\right) e = \frac{1}{|W|}\sum_{w\in W} we = \frac{1}{|W|}\sum_{w\in W} e = \frac{1}{|W|}|W|e = e,$$

so $z \cdot e = e$. Conversely, let $z \cdot e = e$; we want to prove that $w \cdot e = e$ for all $w \in W$. We have

$$z \cdot e = e \implies (wz) \cdot e = e \implies w(z \cdot e) = e \implies w \cdot e = e$$

for all $w \in W$. Hence, $e \in E^W$, which gives $E^1 = E^W$. Therefore, the trace of z is equal to $\dim(E^W)$. □

Extending the base field from $\mathbb{R}$ to $\mathbb{C}$, since every matrix is triangularizable over $\mathbb{C}$. Also, since W is a finite reflection group, for each $w \in W$ there exists m such that $w^m = 1$. Since over $\mathbb{C}$ the minimal polynomial of the operator w consists of distinct linear factors, each w is diagonalisable over $\mathbb{C}$ when viewed as a member of $GL(V)$. Also, we conclude that each eigenvalue is of unit modulus and hence is either 1 or -1, or occurs in a conjugate pair. From this it follows that each w is diagonalisable when it acts on V^* as well. Also, w acts on S_k, where S_k consists of all homogeneous polynomials of degree k along with zero. Let $c_1, c_2, \cdots, c_n$ be the eigenvalues of w when it acts on V. Let t be any complex number, then

$$\det(1 - tw) = (1 - c_1 t)(1 - c_2 t) \cdots (1 - c_n t)$$

and hence

$$\frac{1}{\det(1 - tw)} = \frac{1}{(1 - c_1 t)(1 - c_2 t) \cdots (1 - c_n t)}.$$

Also, we know that $\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots$. Using this identity, we conclude the above equation as

$$\frac{1}{\det(1 - tw)} = (1 + c_1 t + c_1^2 t^2 + \cdots)(1 + c_2 t + c_2^2 t^2 + \cdots) \cdots (1 + c_n t + c_n^2 t^2 + \cdots)$$

and hence

$$\frac{1}{\det(1 - tw)} = \sum_{k \geq 0} \left(\sum_{k_1 + k_2 + \cdots + k_n = k} c_1^{k_1} c_2^{k_2} \cdots c_n^{k_n} \right) t^k.$$

Proposition 3.2.3.

$$\frac{1}{|W|} \sum_{w \in W} \frac{1}{\det(1 - tw)} = \prod_{i=1}^{n} \frac{1}{1 - t^{d_i}},$$

when viewed as a power series in t.

Proof. We know that for a finite reflection group W, any $w \in W$ is diagonalisable when it acts on a finite dimensional vector space V, having eigenvalues of unit modulus. Also, since w is a real matrix, this implies that all the eigenvalues are either 1 or -1, or occur in conjugate pairs. Let $(z_1, z_2, \cdots, z_n)$ be a basis of eigenvectors of w when it acts on V. We know the basis of w when it acts on S_k, which is all monomials of the form $\{z_1^{k_1} z_2^{k_2} \cdots z_n^{k_n} \mid k_1 + k_2 + \cdots + k_n = k\}$.

Next, we claim that all these monomials are eigenvectors of w when it acts on S_k. This is because

$$w(z_1^{k_1} z_2^{k_2} \cdots z_n^{k_n}) = c_1^{k_1} z_1^{k_1} c_2^{k_2} z_2^{k_2} \cdots c_n^{k_n} z_n^{k_n} = (c_1^{k_1} c_2^{k_2} \cdots c_n^{k_n}) z_1^{k_1} z_2^{k_2} \cdots z_n^{k_n}.$$

Hence, each member of the basis is an eigenvector, which implies that w is diagonalisable when it acts on S_k as well, with eigenvalues $(c_1^{k_1} c_2^{k_2} \cdots c_n^{k_n})$ such that $k_1 + k_2 + \cdots + k_n = k$. Thus, the trace of w when it acts on S_k is $\sum_{k_1+k_2+\cdots+k_n=k} c_1^{k_1} c_2^{k_2} \cdots c_n^{k_n}$.

Now, we look at the coefficient of t^k on the LHS:

$$\begin{aligned}\text{coefficient of } t^k \text{ in } \frac{1}{|W|} \sum_{w \in W} \frac{1}{\det(1 - tw)} &= \frac{1}{|W|} \sum_{w \in W} \left(\sum_{k_1+k_2+\cdots+k_n=k} c_1^{k_1} c_2^{k_2} \cdots c_n^{k_n} \right) \\ &= \frac{1}{|W|} \sum_{w \in W} \operatorname{tr}(w).\end{aligned}$$

when w acts on S_k, which is equal to $\operatorname{tr}\left(\frac{1}{|W|} \sum_{w \in W} w\right)$, and this is equal to the dimension of the invariant subspace, denoted by $\dim(R^k)$.

Now, we consider the coefficient of t^k on the RHS, which is the coefficient of t^k in

$$(1+t^{d_1}+\cdots)(1+t^{d_2}+\cdots)\cdots(1+t^{d_n}+\cdots) = \sum_{d_1 k_1 + \cdots + d_n k_n = k} \left((t^{d_1})^{k_1} (t^{d_2})^{k_2} \cdots (t^{d_n})^{k_n} \right).$$

The number of monomials satisfying the condition $d_1 k_1 + \cdots + d_n k_n = k$ is the coefficient of t^k on the RHS. Now, if $f_1, f_2, \cdots, f_n$ is a set of basic invariants of R, then a basis of R^k is $\{f_1^{k_1} f_2^{k_2} \cdots f_n^{k_n}\}$ such that $d_1 k_1 + \cdots + d_n k_n = k$. Thus, the coefficient of t^k on the RHS is the dimension of R^k, proving the proposition. □

Theorem 3.2.4. *Let W be a finite reflection group. Then the order of W is equal to the product of the degrees of the basic invariant polynomials. Moreover, if N is the number of reflections in W, then $\sum_{i=1}^{n} (d_i - 1) = N$.*

Proof. We know that the eigenvalues of a reflection are -1 with multiplicity 1 and 1 with multiplicity $n - 1$. Now we intend to prove that the converse is also true, i.e., if we take any element w of W such that it has $n - 1$ eigenvalues equal to 1 and the remaining eigenvalue is -1, then it must be a reflection. This is because if w has $n - 1$ eigenvalues equal to 1, then it

fixes a hyperplane H pointwise. Also, since w has an eigenvalue -1, there exists a non-zero vector α which is sent to its negative by w. Now, we intend to prove that α is orthogonal to H. Take an arbitrary member of H, say v. Then

$$\langle \alpha, v \rangle = \langle w\alpha, wv \rangle = \langle -\alpha, v \rangle = -\langle \alpha, v \rangle,$$

which implies $\langle \alpha, v \rangle = 0$, and hence each vector of H is orthogonal to α. Therefore $H = H_\alpha$ and $w = s_\alpha$, and hence the converse is also true.

From the above proposition we have,

$$\frac{1}{|W|} \sum_{w \in W} \frac{1}{\det(1 - tw)} = \prod_{i=1}^{n} \frac{1}{1 - t^{d_i}}.$$

Since W is a finite reflection group, every element of W is either a reflection, the identity, or neither a reflection nor the identity. So, if w is the identity then all the eigenvalues of w are 1, and hence $\det(1-tw) = (1-t)^n$. Now, if w is a reflection of W then the eigenvalues of w are $\{-1, 1, \cdots, 1\}$, and hence $\det(1 - tw) = (1 - t)^{n-1}(1 + t)$. If w is neither a reflection nor the identity, then w can have at most $n-2$ eigenvalues equal to 1. This is because if w has $n - 1$ eigenvalues equal to 1, then either all eigenvalues are 1, or $n - 1$ are 1 and one eigenvalue is -1; in the first case w is the identity and in the second w is a reflection, contradicting the assumption. So, it follows that w can have at most $n-2$ eigenvalues equal to 1. Thus $\det(1-tw) = (1-t)^{n-2}h(t)$, where $h(t)$ is a degree 2 polynomial not divisible by $(1 - t)$. Using all these, and multiplying the left hand side by $(1 - t)^n$, we get

$$\frac{1}{|W|} \sum_{w \in W} \frac{(1-t)^n}{\det(1 - tw)} = \frac{1}{|W|} \left(1 + N \frac{1-t}{1+t} + (1-t)^2 h(t) \right),$$

where N is the number of reflections of W.
Multiplying the right hand side by $(1 - t)^n$, we get

$$\prod_{i=1}^{n} \frac{1-t}{1-t^{d_i}} = \prod_{i=1}^{n} \frac{1}{1 + t + t^2 + \cdots + t^{d_i - 1}}.$$

Finally, we get

$$\frac{1}{|W|} \left(1 + N \frac{1-t}{1+t} + (1-t)^2 h(t) \right) = \prod_{i=1}^{n} \frac{1}{1 + t + t^2 + \cdots + t^{d_i - 1}}.$$

Now, putting $t = 1$ in this equation, we get

$$\frac{1}{|W|}(1 + 0 + 0) = \prod_{i=1}^{n} \frac{1}{d_i},$$

and thus

$$|W| = d_1 d_2 \cdots d_n.$$

Now, differentiating the previous equation with respect to t, we get

$$\frac{1}{|W|}\left(N\left(\frac{(-1)(1+t)-(1-t)}{(1+t)^2}\right)+(-2)(1-t)h(t)+(1-t)^2h'(t)\right)=\prod_{i=1}^{n}h_i\sum_{i=1}^{n}\frac{h_i'}{h_i},$$

where $h_i = \frac{1}{1+t+t^2+\cdots+t^{d_i-1}}$. So, we get

$$\begin{aligned}\frac{1}{|W|}&\left(N\left(\frac{(-1)(1+t)-(1-t)}{(1+t)^2}\right)+(-2)(1-t)h(t)+(1-t)^2h'(t)\right)\\&=\prod_{i=1}^{n}\frac{1}{1+t+t^2+\cdots+t^{d_i-1}}\sum\frac{1+2t+3t^2+\cdots+(d_i-1)t^{d_i-2}}{1+t+t^2+\cdots+t^{d_i-1}}.\end{aligned}$$

Putting $t = 1$ in this equation, we get

$$\frac{1}{|W|}\left(N\left(\frac{-2}{4}\right)\right)=\prod_{i=1}^{n}\frac{1}{d_i}\sum\frac{-(1+2+3+\cdots+(d_i-1))}{d_i},$$

$$\implies \qquad \frac{-N}{2}=\sum\frac{(-d_i)(d_i-1)}{2d_i},$$

$$\implies \qquad N=\sum d_i-n \implies \sum d_i = N+n.$$

□

Next, after learning Chevalley's theorem, we intend to prove the converse of the statement, i.e., if G is a finite group acting on S, such that the set of G-invariants in S is generated by n homogeneous algebraically independent elements, then G must be a reflection group. But before that, we will prove a criterion to check the algebraic independence of elements in S.

3.3 Jacobian Criteria to check Algebraic Independence

Theorem 3.3.1. *Let $\{f_1, f_2, \cdots, f_n\}$ be a set of n homogeneous elements in indeterminates $\{x_1, x_2, \cdots, x_n\}$. Define J to be the determinant of the matrix $\left(\frac{\partial f_i}{\partial x_k}\right)$. Then the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically independent if and only if $J \neq 0$.*

Proof. First, we assume that $J \neq 0$. We intend to prove that the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically independent. Suppose, for contradiction, that the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically dependent. It implies that there exists a polynomial $h(y_1, y_2, \cdots, y_n) \neq 0$ such that $h(f_1, f_2, \cdots, f_n) = 0$. Choose h of minimal degree. Differentiating h with respect to x_k, we get

$$\sum_{j=1}^{n} \frac{\partial h}{\partial y_j} \frac{\partial f_j}{\partial x_k} = 0.$$

We can see this as n equations in n variables where the coefficients are $\frac{\partial f_j}{\partial x_k}$. Now we intend to prove that this matrix has determinant 0. This is because since h is non-zero, there exists i such that $\frac{\partial h}{\partial y_i} \neq 0$. Now, $\frac{\partial h}{\partial y_i}(f_1, f_2, \cdots, f_n) \neq 0$ as $\deg\left(\frac{\partial h}{\partial y_i}\right) < \deg(h)$. So, these n equations have a non-zero solution, and hence the coefficient matrix has determinant zero. Thus $J = 0$, which contradicts the assumption. Hence, the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically independent.
Conversely, we now assume that the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically independent, and we intend to prove that $J \neq 0$. We know that the field of fractions of R has transcendence degree n, which implies that R has exactly n algebraically independent elements. Hence, if we take the set $\{x_i, f_1, f_2, \cdots, f_n\}$, then this set is definitely algebraically dependent, meaning there exists $h_i(y_0, y_1, y_2, \cdots, y_n) \neq 0$ such that $h_i(x_i, f_1, f_2, \cdots, f_n) = 0$ for each i. Choose such h_i of minimal degree. Differentiating with respect to x_k, we get

$$\sum_{j=1}^{n} \frac{\partial h_i}{\partial y_j}(x_i, f_1, \cdots, f_n) \frac{\partial f_j}{\partial x_k} + \frac{\partial h_i}{\partial y_0}(x_i, f_1, \cdots, f_n) \frac{\partial x_i}{\partial x_k} = 0,$$

which implies

$$\sum_{j=1}^{n} \frac{\partial h_i}{\partial y_j}(x_i, f_1, \cdots, f_n) \frac{\partial f_j}{\partial x_k} = -\frac{\partial h_i}{\partial y_0}(x_i, f_1, \cdots, f_n) \frac{\partial x_i}{\partial x_k}.$$

Now, viewing this equation in terms of matrices, we see that the right hand side is a diagonal matrix with non-zero entries and hence non-zero determinant. All terms are non-zero because h_i was chosen of minimal degree. It follows that $J \neq 0$.Conversely, we now assume that the set $\{f_1, f_2, \cdots, f_n\}$ is algebraically independent, and we intend to prove that $J \neq 0$. We know that the field of fractions of R has transcendence degree n, which implies that R has exactly n algebraically independent elements. Hence, if we take the set

$\{x_i, f_1, f_2, \cdots, f_n\}$, then this set is definitely algebraically dependent, meaning there exists $h_i(y_0, y_1, y_2, \cdots, y_n) \neq 0$ such that $h_i(x_i, f_1, f_2, \cdots, f_n) = 0$ for each i. Choose such h_i of minimal degree. Differentiating with respect to x_k, we get

$$\sum_{j=1}^{n} \frac{\partial h_i}{\partial y_j}(x_i, f_1, \cdots, f_n)\frac{\partial f_j}{\partial x_k} + \frac{\partial h_i}{\partial y_0}(x_i, f_1, \cdots, f_n)\frac{\partial x_i}{\partial x_k} = 0,$$

which implies

$$\sum_{j=1}^{n} \frac{\partial h_i}{\partial y_j}(x_i, f_1, \cdots, f_n)\frac{\partial f_j}{\partial x_k} = -\frac{\partial h_i}{\partial y_0}(x_i, f_1, \cdots, f_n)\frac{\partial x_i}{\partial x_k}.$$

Now, viewing this equation in terms of matrices, we see that the right hand side is a diagonal matrix with non-zero entries and hence non-zero determinant. All terms are non-zero because h_i was chosen of minimal degree. It follows that $J \neq 0$. □

Corollary 3.3.2. *If J is the Jacobian of $\{f_1, f_2, \cdots, f_n\}$, which are n homogeneous algebraically independent elements, then the degree of J is the number of reflections in W.*

Proof. Since from above we know that $J \neq 0$, and using the definition of the determinant, we know that

$$J = \sum_{\pi} \pm \prod_{i=1}^{n} \frac{\partial f_i}{\partial x_{\pi(i)}}.$$

Hence the degree of J is $\sum(d_i - 1)$, where d_i are the degrees of f_i. From above we know that $\sum(d_i - 1) = N$, where N is the number of reflections in W. □

3.4 Converse of Chevalley's Theorem

Theorem 3.4.1. *Let G be a finite group acting on a finite-dimensional Euclidean space V over $\mathbb{R}$, and let G act naturally on $S = \mathbb{R}[x_1, x_2, \ldots, x_n]$. Suppose that S^G is generated by n homogeneous algebraically independent elements of positive degree. Then G is a finite reflection group.*

Proof. Let H be the subgroup of G generated by all reflections in G. We wish to show that $H = G$. It suffices to show that $|H| = |G|$, since $H \subseteq G$.

By hypothesis, S^G is generated by n homogeneous algebraically independent polynomials of positive degree; call them $g_1, g_2, \ldots, g_n$ with $\deg(g_i) = e_i$.

Since H is itself a finite reflection group (being generated by reflections), by Chevalley's theorem, S^H is also generated by n homogeneous algebraically independent elements of positive degree; call them $f_1, f_2, \ldots, f_n$ with $\deg(f_i) = d_i$.

Since $H \subseteq G$, we have $S^G \subseteq S^H$, so each g_i can be expressed as a polynomial in $f_1, f_2, \ldots, f_n$:

$$g_i = g_i(f_1, f_2, \ldots, f_n), \quad i = 1, 2, \ldots, n.$$

Differentiating with respect to x_k, we obtain:

$$\frac{\partial g_i}{\partial x_k} = \sum_{j=1}^{n} \frac{\partial g_i}{\partial f_j} \cdot \frac{\partial f_j}{\partial x_k}.$$

In matrix form, this reads:

$$J(g_1, \ldots, g_n) = \left(\frac{\partial g_i}{\partial f_j} \right) \cdot J(f_1, \ldots, f_n),$$

where $J(g_1, \ldots, g_n)$ and $J(f_1, \ldots, f_n)$ denote the respective Jacobian matrices with respect to $(x_1, \ldots, x_n)$. By the Jacobian criterion, $\det J(g_1, \ldots, g_n) \neq 0$ and $\det J(f_1, \ldots, f_n) \neq 0$, and therefore:

$$\det \left(\frac{\partial g_i}{\partial f_j} \right) \neq 0.$$

This means:

$$\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} \frac{\partial g_i}{\partial f_{\sigma(i)}} \neq 0,$$

so there exists a permutation σ such that $\prod_{i=1}^{n} \frac{\partial g_i}{\partial f_{\sigma(i)}} \neq 0$. After reindexing the f_i, we may assume:

$$\prod_{i=1}^{n} \frac{\partial g_i}{\partial f_i} \neq 0.$$

In particular, $\frac{\partial g_i}{\partial f_i} \neq 0$ for each i, which means f_i actually appears in g_i. By degree considerations, this forces $e_i \geq d_i$ for all i.

On the other hand, since $H \subseteq G$, the set of reflections in H is exactly the set of reflections in G (as H is generated by all reflections of G). By the degree formula:

$$|G| = \prod_{i=1}^{n} e_i, \qquad |H| = \prod_{i=1}^{n} d_i,$$

and:

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} d_i,$$

since both equal $n + |\{\text{reflections in } G\}|$. Combined with $e_i \geq d_i$ for all i, we conclude $e_i = d_i$ for all i, and hence:

$$|G| = \prod_{i=1}^{n} e_i = \prod_{i=1}^{n} d_i = |H|.$$

Since $H \subseteq G$ and $|H| = |G|$, we conclude $H = G$. Therefore, G is a finite reflection group. □

Remark

Question: Let W be a finite reflection group acting on S, and let S^G denote the set of all invariant polynomials. If $\{g_1, g_2, \ldots, g_n\}$ is a set of n homogeneous algebraically independent elements in $R = S^G$, will they necessarily generate R?

Answer: No.

Example 3.4.1 (Counter example)**.** *Take $W = S_2$ and $V = \mathbb{R}^2$. Consider the following two sets:*

$$A = \{xy,\ x + y\}, \qquad B = \{x^2 + y^2,\ x^2y^2\}.$$

Both sets consist of homogeneous algebraically independent elements. However, A generates S^G, whereas B does not generate S^G. This is because xy can never be obtained from the elements of B, and hence $xy \notin$ span of B as a polynomial algebra. That A generates S^G will be proved in the next theorem.

Remark 3.4.2. *Since the answer is no, one may ask: what additional condition is required to guarantee that $\{g_1, g_2, \ldots, g_n\}$ generates R? The required extra condition is:*

$$|W| = \prod_{i=1}^{n} e_i,$$

where $e_i = \deg(g_i)$ for each i.

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